OSDP 3.0 Proposed PIV Enhancements

Security Industry Association

September 2, 2026

Contents

Changelog

2026-08-12 Proposed Secure PIN Entry changes and consistency cleanup:

- Proposed clear, mode-specific SPE behavior for status, PIN acquisition and verification, cached-PIN verification, cache/abort, and attempts-remaining queries. The proposal also defines finite default timeouts, cache and PIN-verified lifecycles, and ACK/poll completion behavior.
- Proposed a one-byte `osdp_PIVSPER` reply whose value is interpreted by the requested mode. The OSDP 2.2 `osdp_PR01SPER` reply remains unchanged.
- Standardized Enhanced PIV names, including renaming the autonomous PIV Auto reply to `osdp_PIVSTATUS`. Corrected tentative code conflicts by assigning `osdp_PIVSELECT` to `0xB0` and `osdp_PDERROR` to `0x8A`.

2026-06-18 PIVSELECT, PDERROR, and VCI consistency updates:

- Added `osdp_PIVSELECT` as an OSDP-native direct AID selection command using tentative command code `0xA7`; clarified its ISO/IEC 7816 `SELECT by DF name` mapping, selected-AID status update, and application-state clearing rules.
- Removed stale `osdp_CRAUTHR` `osdp_PDERROR` wording so codes `0x0006`/`0x0007` retain their common meanings for secure-channel and multi-part errors.
- Clarified that VCI pairing codes are transmitted from the ACU to the PD only over OSDP Secure Channel 2.0 and are used by the PD to establish VCI secure messaging with the credential.
- Tightened and clarified verbiage by replacing imprecise “legacy” terminology throughout normative text with neutral wording such as unchanged, OSDP 2.2-format, and poll-based; retained “legacy” only for withdrawn algorithms such as RSA-1024 test material.

2026-05-30 Added source-backed VCI and PIV Auto examples, clarified PIV Auto interoperability requirements, and improved reviewability:

- Added NIST-derived VCI and PIV APDU fixture coverage, GSA-backed demonstration keys, and reproducible worked examples for PIV Auto and VCI validation. RSA-1024 material is retained only as informative legacy/non-current test-vector material.
- Clarified PIV Auto cryptographic inputs and response handling, including KMAC-derived challenge construction, RSA and ECC `GENERAL AUTHENTICATE` templates, nonzero UUID requirements, and complete returned `0x7C` Dynamic Authentication Templates in `osdp_PIVSTATUS`.
- Aligned related command semantics for capability advertising, `osdp_CARDSTATUS` errors, selective `osdp_PIVDATA` replies, `osdp_PIVSPE` status/abort modes, and VCI trust-anchor inventory.
- Reworked Appendix C and the references section for standards review: examples are explicitly informative, vector provenance is documented, source fixtures are framed as public test material, and normative references now include the needed APDU, KMAC, X.509, and requirement-language specifications.
- Added supporting scripts and tests for deterministic simulation, fixture replay, VCI import/validation, Appendix C generation, YubiKey material loading, live APDU capture, and reproducible PDF/table builds.

2026-05-27 Version renamed from OSDP 2.3 to OSDP 3.0 throughout. Consolidated PIV Auto, biometric-result, VCI trust-anchor, and SPE updates:

- Removed `osdp_PIVSPE` flag `0x01` (retry continuation); each `osdp_PIVSPE` command now initiates exactly one PIN-entry session and `osdp_PIVSPER` terminates the session unconditionally, including on PIN mismatch. ACUs wishing to allow additional attempts **MAY** re-issue `osdp_PIVSPE`, using the *Attempts Remaining* count from the preceding `osdp_PIVSPER` to guide retry policy. Bit `0x01` is now reserved.
- Aligned `osdp_PIVSPER` and `osdp_PIVBIOMATCHR` result codes for credential removal, changed `osdp_PIVBIOMATCHR 0x03` to PIN verification required, reserved `osdp_PIVBIOMATCHR 0x07`, clarified biometric assurance limits, and specified that only off-card biometric match success returns the 32-byte SHA-256 hash of tag `0xBC`.
- Clarified the Function Code 14 biometric-support reference for `osdp_PIVLOADTPK` and added CAC as PIV mode data model `0x03`.
- Added PIV Auto KMAC challenge derivation, required series and sequence counters, fixed-width supplemental entropy, and the `osdp_PIVSTATUS` poll reply for reporting credential identity and signed challenge responses. PIV Auto is configured through `osdp_PIVMODE` and is restricted to Secure Channel 2.0 sessions that are not using factory keys.
- Replaced the SP 800-73 Intermediate CVC anchor model with an OSDP-defined VCI trust anchor record (tag `0x7F50`) carrying a Profile Identifier (`0x80` = `0x01`), an eight-byte Issuer Identifier Number (`0x42`), and a DER-encoded SubjectPublicKeyInfo (`0x7F49`). Maximum conforming record size is 576 bytes (`0x0240`), derived from the RSA-4096 key case.
- Added normative IIN derivation rule: the IIN in `osdp_PIVVCILOADTA` and in the trust anchor record **SHALL** be the leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate.
- Tightened `osdp_PIVVCILOADTA` multi-part encoding: Anchor Length and MpSizeTotal ranges corrected to `0x0000`-`0x0240`; added normative constraint that `MpOffset + MpFragmentSize` **SHALL NOT** exceed `Anchor Length`; PD **SHALL** verify the reassembled byte string is exactly *Anchor Length* bytes and a single `0x7F50` record before committing.
- Documented variable-length PD storage accounting: PDs **SHALL** account for anchors by their actual *Anchor Length*, not a fixed maximum; replacement is atomic (prior anchor remains if replacement fails); delete releases actual stored bytes.
- Specified the two-tier PD verification path during VCI establishment: direct (trust anchor IIN matches secure messaging CVC IIN, anchor SPKI verifies CVC signature) and intermediate (no direct match triggers retrieval of PIV data object `0x1017/5FC122`; Intermediate CVC Subject Identifier must equal CVC IIN, Role `0x5F4C` must be `0x12`, anchor IIN must match Intermediate CVC IIN, anchor SPKI verifies Intermediate CVC, Intermediate CVC verifies secure messaging CVC). PDs **SHALL NOT** perform X.509 path validation or revocation for VCI trust anchors.
- Added normative ACU post-VCI validation requirements: after `osdp_CARDSTATUSR` reports VCI established, the ACU **SHALL** obtain the content-signing certificate (from cache or via `osdp_PIVDATA` for `0x1017`/inner tag `0x70`), verify SKI matches *VCI Anchor IIN*, verify SPKI

matches the stored anchor record, verify PIV or PIV-I content-signing EKU OID, verify validity period (expired certificate **SHALL** cause session rejection and anchor unload), and perform X.509 path validation and revocation per deployment policy.

- Restored *VCI Trust Anchor ID* to `osdp_CARDSTATUSR` with normative language; added *VCI CVC IIN* and *VCI Anchor IIN* fields (each 8 bytes) so the ACU receives the IIN pair needed to distinguish direct and intermediate verification paths.
- Updated `osdp_PIVVCITASTATUSR`: clarified that *Available Bytes* reflects variable-length accounting and that the per-record IIN is the anchor selector from the stored trust anchor record, not necessarily the secure messaging CVC IIN reported in `osdp_CARDSTATUSR`.
- Added VCI security preconditions: all VCI-related commands **SHALL** be rejected unless OSDP Secure Channel 2.0 is active and not using factory/default keys; failures reported via `osdp_PDERROR` `0x0006`.

2026-01-21 Error-handling cleanup and code assignments: **(1)** `osdp_PIVPUTDATA` now **SHALL** report failures via `osdp_PDERROR` (status word optional) so writes can't silently succeed. **(2)** `osdp_PIVSPE` adds result code `0x09` in `osdp_PIVSPER` for "VCI not established." **(3)** PDERROR is credential/global only; PD capability gaps stay `osdp_NAK`. **(4)** OSDP 2.2-format replies (e.g., `osdp_CRAUTHR`, `osdp_PIVDATAR`) remain unchanged, with optional follow-up `osdp_PDERROR` detail if the PD has it. **(5)** Documented tentative command/reply codes for the new PIV set: commands `0xA6` (VCI trust-anchor status) through `0xAF` (VCI trust-anchor load), and replies `0x85`-`0x89` for the new responses.

2026-01-07 Added a PIV Mode interface-mask table aligned to Function Code 13 bits, clarified `osdp_CRAUTHR` reply structure (full GENERAL AUTHENTICATE response without SW1|SW2; status words reported via `osdp_PDERROR` only), tightened `osdp_CARDSTATUS`/`osdp_CARDSTATUSR` behavior (solicited-only after PD preparation; error on credential removal), remapped the PIV error codes to the `0x10xx` block per subcommittee guidance, added PIV APDU references, aligned `osdp_PIVDATA` tag selection to allow Discovery (`0x7E`)/BIT (`0x7F61`) subtags in line with SP 800-73-4, clarified that `osdp_PDERROR` conveys a single error per reply, and clarified `osdp_PIVXMITPAIRING` handling so pairing codes can be provided before VCI is established when the credential requires them.

2025-12-10 Consolidated application-level errors on `osdp_PDERROR` (two-byte codes, optional SW1|SW2 for APDU failures, multi-error support); moved the PIV error set to `0x0100`-`0x01FF` and updated PIV commands to use `osdp_PDERROR` (transport faults remain `osdp_NAK`); made the Global Interface Mask in `osdp_PIVMODE` a required byte (no TLV tag) with an updated field layout; clarified that `osdp_CARDSTATUSR` is solicited-only after PD preparation with poll-based `osdp_RAW` as the aggregated card-present indication; and noted the deprecation of `osdp_GENAUTH` with a recommendation to migrate to `osdp_CRAUTH`.

2025-12-04 Clarified that `osdp_CRAUTHR` returns the full `GENERAL AUTHENTICATE` data including the Dynamic Authentication Template (excluding only the ISO/IEC 7816 status word).

2025-11-12 Added PIV Biometric Match request/reply and tables (one-bit type values only); moved Enhanced PIV NAKs to a 0x20+ base (added 0x27 Credential Not Present/Removed, 0x28 Biometric Type

Not Supported; clarified 0x23 Credential Security Status Not Satisfied); aligned CRAUTHR error handling to reference the enhanced NAK set; relocated Enhanced PIV (0x04) and PIV Auto (0x08) indicators to Function Code 12 and cleaned Function Code 13 to interface-only bits (removed non-interface flags; removed PIV Auto from Function Code 14); updated CARDSTATUS gating to Function Code 12 bit 0x04; added a PIV Mode Flags table and clarified PIV Mode AID encoding; enabled PDF TOC hyperlinks and improved table/caption pagination.

2025-10-30 Clarified CRAUTH command/reply behaviour (including RSA, AES, and ECC support), overhauled PIV Mode configuration with TLV-based application and interface controls (including PIV Auto), added card status request/reply messages, expanded PIV data retrieval semantics, introduced PIV Put Data, PIV Secure PIN Entry, TWIC privacy-key load, VCI pairing transmit, and VCI trust-anchor load/status commands, and documented the enhanced error-code set (including VCI/PIV Auto conditions).

2025-09-17 Major PIV Mode and SPE revisions.

2025-06-25 Added multi-AID usage and VCI support notes.

2025-05-27 New function code definitions for 3.0 biometric handling.

1 Introduction

This document captures proposed enhancements to the SIA Open Supervised Device Protocol (OSDP) needed to carry FICAM-compliant Personal Identity Verification (PIV) workflows. The material consolidates draft content, reviewer comments, and interoperability feedback collected during the 2025 working sessions and re-expresses it as normative guidance for command authors and device implementers.

The chapters that follow highlight changes required for existing OSDP 2.2 commands, introduce new transactions where no equivalent existed, and provide supplemental examples that reflect current federal credentialing practices. Wherever the proposal diverges from the baseline specification, the text records both the behavioral requirements and the error handling expectations so that access control units (ACUs) and peripheral devices (PDs) can adopt the profile without relying on vendor extensions.

Backward compatibility: when a PD encounters a credential-side error on an OSDP 2.2-format reply (for example `osdp_CRAUTHR` or `osdp_PIVDATAR`), it continues to return the unchanged reply shape (often with `DATA_LEN = 0`) and may optionally send an `osdp_PDERROR` with additional detail on a subsequent `osdp_POLL`. ACUs are expected to tolerate or ignore the optional `osdp_PDERROR`; OSDP 2.2 ACUs may log the unknown reply code and proceed.

2 Changes to OSDP 2.2

This section documents the normative deltas that apply when an OSDP 2.2 implementation adopts the Enhanced PIV profile. Each subsection references the original command definition and records the precise behavioral updates, encoding clarifications, and new error handling rules introduced by this proposal.

2.1 Updated Command Definitions

2.1.1 Authentication Challenge (osdp_CRAUTH)

OSDP 2.2 transports the ISO/IEC 7816-4 `GENERALAUTHENTICATE` command through `osdp_CRAUTH`. The OSDP 3.0 profile maintains the same framing fields (`TOTAL`, `OFFSET`, `DATA_LEN`) but clarifies how the command payload maps to the PIV operation selected by the algorithm and key reference.

- The *Algorithm* field continues to be sent in the first fragment only. Codes `0x05`, `0x06`, and `0x07` select RSA private-key operations and preserve backwards compatibility with OSDP 2.2. Codes `0x08`, `0x0A`, and `0x0C` select AES encryption using 128-, 192-, and 256-bit keys respectively. Codes `0x11` and `0x14` identify ECC P-256 and P-384 mechanisms; the selected PIV key reference determines whether the mechanism is used for ECDSA signature generation or ECDH key establishment.
- The *Key* field retains its role as the card's reference identifier. PDs **SHALL NOT** reinterpret or remap the identifier.
- For RSA algorithms (`0x05`, `0x06`, `0x07`), the *Challenge* field carries the encoded datum for the RSA private-key operation. The PD forwards the value unmodified; padding and digest encoding are the responsibility of the ACU.
- For AES algorithms (`0x08`, `0x0A`, `0x0C`), the *Challenge* field contains the plaintext block(s) to be encrypted. PDs **SHALL** send the data to the credential without padding; ACUs **SHALL** provide data aligned to the block size dictated by the selected key length and card profile.
- For ECC signature operations using key references `0x9A`, `0x9C`, or `0x9E` with algorithms `0x11` or `0x14`, the *Challenge* field carries the pre-hashed input to be signed. The PD **SHALL** issue `GENERALAUTHENTICATE` with command data encoded as `7C Len 82 00 81 Len Challenge`; the credential returns the ECDSA signature in tag `0x82`. Exponentiation tag `0x85` **SHALL NOT** be used with `0x9A`, `0x9C`, or `0x9E`.
- For ECC key-establishment operations using key reference `0x9D` or retired key-management references, algorithms `0x11` and `0x14` use the *Challenge* field to convey the peer public key. The PD **SHALL** issue `GENERALAUTHENTICATE` with command data encoded as `7C Len 82 00 85 Len (04 || X || Y)`, where `X` and `Y` are the uncompressed coordinates for the selected curve and are left-padded with `0x00` as required. The credential performs elliptic-curve Diffie-Hellman in accordance with SP 800-73.

Table 1: Authentication Challenge (osdp_CRAUTH)

Size	Name	Meaning	`Value`
1	CMND	Command identifier.	`0xA5`
2	TOTAL	Length of the complete message, least-significant byte first.	`0x0000–0xFFFF`
2	OFFSET	Offset of this fragment within the complete message, least-significant byte first.	`0x0000–0xFFFF`
2	DATA_LEN	Length of the fragment payload, least-significant byte first.	`0x0000–0xFFFF`
1	Algorithm	Cryptographic mechanism. Supported values include `0x05`/`0x06`/`0x07` (RSA private-key operation), `0x08`/`0x0A`/`0x0C` (AES encryption with 128/192/256-bit keys), and `0x11`/`0x14` (ECC P-256/P-384; operation depends on the key reference).	`0x00–0xFF`
1	Reference Identifier	Key reference passed to `GENERALAUTHENTICATE`.	`0x00–0xFF`
0-n	Challenge	For RSA algorithms, the encoded datum for the private-key operation. For AES algorithms, the plaintext block(s) to be encrypted, pre-padded by the ACU as required. For ECC `0x9A`/`0x9C`/`0x9E`, the pre-hashed signature input in tag `0x81`; for ECC `0x9D`/retired key-management references, the peer public key in tag `0x85`.	`0x00–0xFF`

Deprecated osdp_GENAUTH The `osdp_GENAUTH` command is deprecated and expected to be removed in a future revision of this specification. Using a PIV credential to decrypt or process arbitrary data is unsafe and unnecessary for credential verification; implementations **SHOULD** use `osdp_CRAUTH` flows instead.

2.1.2 Response to Challenge (osdp_CRAUTHR)

The reply continues to use the OSDP 2.2 multi-part format. Successful execution yields a response containing the credential's entire `GENERALAUTHENTICATE` data (minus the ISO/IEC 7816 status word), including the returned Dynamic Authentication Template; PDs **SHALL NOT** strip outer BER-TLV tags or lengths. If the credential reports an error, the PD **SHALL** return the unchanged `osdp_CRAUTHR` reply (for example with `DATA_LEN = 0`) to preserve OSDP 2.2 behaviour. When available, the PD **SHOULD** also return osdp_PDERROR on a subsequent osdp_POLL with an applicable PIV code—for example 0x1020 (function not supported by credential), 0x1022 (insufficient credential storage), 0x1023 (credential security status not satisfied, including PIN-required conditions), 0x1024 (data object/tag not found), 0x1025 (VCI not established), or 0x1027 (credential not present/removed). Generic parameter errors may continue to use 0x0005 (incorrect parameter). PIN-entry outcomes are reported through command-specific result paths, such as osdp_PIVSPER, when secure PIN entry is used. ACUs **SHALL** tolerate an optional osdp_PDERROR following that reply; OSDP 2.2 ACUs may log and ignore the unknown reply code.

Reply structure

osdp_CRAUTHR **SHALL** return the credential's full ISO/IEC 7816 GENERALAUTHENTICATE response payload, including the Dynamic Authentication Template, framed by the standard OSDP multi-part fields. PDs **SHALL NOT** include SW1|SW2 in osdp_CRAUTHR; any status words from the credential **SHALL** be carried only in osdp_PDERROR when a card/APDU failure occurs. PDs **SHALL NOT** strip outer BER-TLV tags or lengths from the returned payload.

Table 2: Response to Challenge (osdp_CRAUTHR)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	0x82
2	TOTAL	Length of the complete response, least-significant byte first.	0x0000–0xFFFF
2	OFFSET	Offset of this fragment within the complete response, least-significant byte first.	0x0000–0xFFFF
2	DATA_LEN	Length of the fragment payload, least-significant byte first.	0x0000–0xFFFF
0-n	Response Data	Signature, encrypted block, or shared-secret material returned by the credential. The credential's BER-TLV structure is preserved exactly as received (status word excluded).	0x00–0xFF

2.1.3 Get PIV Data (osdp_PIVDATA)

OSDP 2.2 defined osdp_PIVDATA with a five-byte payload that selected a PIV object and optional element. The original text did not explicitly specify how to pad identifiers shorter than three bytes, how nested tags should be returned, or how to interpret the data offset. OSDP 3.0 clarifies these behaviors so that ACUs and PDs apply a single normative approach.

- The ACU **SHALL** transmit the PIV Object Identifier as three bytes encoded most-significant byte first. When the identifier contains fewer than three significant bytes, the ACU **SHALL** left-pad the value with 0x00 or 0x0000, as required, so that the transmitted field is exactly three bytes. The PD **SHALL**

NOT forward those pad bytes to the credential; it **SHALL** issue the ISO 7816-4 `GET DATA` APDU using only the significant identifier bytes. If VCI secure messaging is established, the PD **SHALL** use the appropriate secure channel automatically; no explicit selector is present in the command.

- The field previously described as "PIV element ID" is renamed *PIV Tag*. A tag value of `0x00` directs the PD to return the complete BER-TLV structure, including the native outer tag (for example `0x7E` Discovery, `0x7F61` BIT Group Template) or the `0x53`/`0x73` wrapper used by other PIV objects. Any non-zero value selects a single root-level tag within the object. When the outer tag is `0x53` or `0x73`, the PD **SHALL** strip that wrapper before returning the selected tag. For other outer tags, the PD **SHALL** return the requested tag if present and **SHALL NOT** reject the request solely because the wrapper is not `0x53`/`0x73`. Tags **SHALL** be a single byte in the range `0x01`-`0xFF`; multi-byte DER tags are not supported and **MUST** be retrieved by issuing a tag value of `0x00` and parsing the returned object at the ACU.

- The data offset now supports one- or two-byte encodings. PDs **SHALL** accept either form and determine which was used based on the command length. The offset applies to the selected payload (entire object when the tag is zero, or the chosen tag otherwise). When a credential-side error occurs, PDs **SHALL** return the unchanged `osdp_PIVDATAR` reply (for example with `DATA_LEN = 0`) to maintain OSDP 2.2 compatibility. If available, PDs **MAY** report additional detail via `osdp_PDERROR` on a subsequent `osdp_POLL` using code `0x1023` when security prerequisites are not satisfied, `0x1024` when the requested object or tag is absent, or `0x1027` when the credential is not present or is removed; see Subsection 2.2. ACUs **SHALL** tolerate an optional `osdp_PDERROR` following that reply; OSDP 2.2 ACUs may log and ignore the unknown reply code.

Table 3: Get PIV Data (osdp_PIVDATA)

Size	Name	Meaning	`Value`
1	CMND	Command identifier.	`0xA3`
3	PIV Object Identifier	Three-byte identifier of the PIV data object per SP 800-73-4, encoded MSB first. Leading zero bytes SHALL be supplied by the ACU when the identifier is shorter than three bytes; the PD SHALL drop those pad bytes before issuing the APDU.	`0x00-0xFF`[1]
1	PIV Tag	Single-byte ASN.1 DER tag of a root-level element. Tag value `0x00` selects the entire object; non-zero tags select a root-level element and require the PD to strip the outer `0x53`/`0x73` wrapper, verify the length, and return only the tagged value. Multi-byte tags are not directly addressable and SHALL be retrieved by requesting tag `0x00`.	`0x00; or 0x01-0xFF`

Size	Name	Meaning	Value
1-2	Data Offset	Byte offset within the requested content. Single-byte and two-byte encodings SHALL be accepted; the PD determines the format from the packet length.	`0x00–0xFFFF`

[1] Each position MAY carry any value from `0x00` to `0xFF`; the ACU pads on the left with `0x00` when the identifier has fewer than three significant bytes.

2.1.4 PIV Data Reply (osdp_PIVDATAR)

The reply structure is unchanged from a field-count perspective, but OSDP 3.0 aligns the semantics with the command refinements above:

- `TOTAL`, `OFFSET`, and `DATA_LEN` continue to be encoded LSB first; they now explicitly report the size, fragment base, and fragment length for the content selected by the `osdp_PIVDATA` request (entire object or chosen tag).
- `CARD_DATA` **SHALL** convey the exact bytes obtained from the credential after applying any tag stripping specified by the command. No additional framing or padding is permitted.

Table 4: PIV Data Reply (osdp_PIVDATAR)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	`0x80`
2	Total Length	Length in bytes of the complete data object or tagged element, least-significant byte first.	`0x0000–0xFFFF`
2	Fragment Offset	Offset in bytes of this fragment within the complete payload, least-significant byte first.	`0x0000–0xFFFF`
2	Fragment Length	Length in bytes of the data carried in this fragment, least-significant byte first.	`0x0000–0xFFFF`
0-n	Card Data	Requested data bytes exactly as returned by the credential. When a tagged element was selected, this sequence contains the complete selected BER-TLV, including tag, length, and value.	`0x00–0xFF`

2.2 PD Error Codes for PIV

PD error codes are carried in `osdp_PDERROR` using two-byte codes (least-significant byte first). The common PD codes remain available (for example `0x0001` SC_ERROR with optional SW1|SW2, `0x0002` SC_TIMEOUT, `0x0004` TLV_MISSING, `0x0005` INVALID_PARAM, `0x0006` SECURE_CHANNEL_REQUIRED, `0x0007` MP_INVALID, `0x0008` MP_OUT_OF_SEQUENCE, `0x0009` MP_OUT_OF_BOUNDS). The following PIV-specific codes occupy the `0x1000`-`0x10FF` range:

Table 5: Enhanced PIV Error Codes

Code	Description
0x1020	Function not supported by credential (PD capability gaps **SHALL** use `osdp_NAK` function-not-supported).
0x1021	Insufficient PD memory or storage to complete the requested operation.
0x1022	Insufficient credential memory or storage space.
0x1023	Credential security status not satisfied.
0x1024	Data object or tag not found.
0x1025	VCI not established.
0x1027	Credential not present or removed.

2.2.1 PD Application Error Reply (osdp_PDERROR)

PDs **SHALL** use `osdp_PDERROR` only for credential-side or other global application errors that are not already defined in a command-specific reply. The error code is two bytes, encoded least-significant byte first. When a card/APDU failure occurs (for example SC_ERROR or a PIV APDU failure), the PD **MAY** append the ISO/IEC 7816 status word (SW1|SW2) when it is available to the application layer; otherwise the status word is omitted. A single error is reported per `osdp_PDERROR` reply. Transport-level faults (framing, CRC, sequence) continue to use `osdp_NAK`.

Table 6: PD Application Error Reply (osdp_PDERROR)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	0x8A (tentative)
2	Error Code	Two-byte error code, least-significant byte first.	0x0001–0x10FF
0 or 2	Status Word	Optional ISO/IEC 7816 status word for card/APDU failures; present only when the PD can surface it to the application layer.	–

2.3 Updated Function Codes

2.3.1 Function Code 12 – Smart Card Support

Function Code 12 advertises smart-card transport capabilities and profile support at the PD. The compliance byte uses the following bit assignments:

Table 7: Function Code 12 Compliance Flags

`Mask`	Description
`0x01`	PD supports transparent reader mode.
`0x02`	PD supports extended packet mode.
`0x04`	PD supports the OSDP 3.0 Enhanced PIV profile, including `osdp_PIVDATA` refinements and related commands.
`0x08`	PD supports OSDP 3.0 autonomous PIV operation (PIV Auto).

Bits not listed are reserved and **SHALL** be transmitted as zero.

2.3.2 Function Code 13 – Reader Interfaces

OSDP 2.2 defined Function Code 13 as a reader count indicator with a compliance byte fixed at `0x00`. Enhanced PIV-capable PDs **SHALL** continue to populate the reader count field, but they MAY set bits within the compliance byte to advertise the downstream credential technologies they expose. A value of `0x00` remains valid and retains the original OSDP 2.2 meaning of “interface presence unspecified.”

When a PD reports specific interfaces, it **SHALL** encode the compliance byte using the flags in Table 8. The reader count byte continues to report the number of attached credential interfaces.

Table 8: Function Code 13 Compliance Flags

`Mask`	Description
`0x01`	PD presents a contact card interface (ISO 7816 or equivalent).
`0x02`	PD presents a high-frequency contactless credential interface (13.56 MHz NFC).
`0x04`	PD presents a low-frequency credential interface (approximately 125 kHz prox).
`0x08`	PD supports Virtual Contact Interface (VCI) secure messaging for applicable modes.
`0x10`	PD presents a Bluetooth credential interface.
`0x20`	PD presents a barcode credential interface.
`0x40`	PD presents an ultra-high-frequency credential interface (UHF/RAIN RFID).

All other bits are reserved for future use and **SHALL** be transmitted as zero.

2.3.3 Function Code 14 – Biometric Support

Function Code 14 remains backward compatible with OSDP 2.2. A compliance value of `0x00`, `0x01`, `0x02`, or `0x03` retains its original meaning and indicates that the PD does not support the OSDP 3.0 biometric command set. Devices that support the enhanced commands **SHALL** set Bit 7 (`0x80`). When Bit 7 is asserted, the remaining bits of the compliance byte are interpreted as a capability bitfield. Bits not explicitly defined in Table 9 are reserved and **SHALL** be transmitted as zero.

Table 9: Function Code 14 Compliance Flags

Mask	Description
0x80	PD supports the OSDP 3.0 biometric command set.
0x01	PD performs off-card fingerprint comparison using FIPS 201 BIO ANSI INCITS 378 templates.
0x02	PD performs on-card fingerprint comparison using FIPS 201 OCC processes.
0x04	PD performs off-card iris comparison aligned with FIPS 201 BIO guidance.
0x08	PD performs off-card facial comparison using ANSI INCITS 385-2004 images stored on the credential.

3 New or Enhanced PIV Commands

This section introduces commands defined exclusively for the OSDP 3.0 Enhanced PIV profile.

3.1 PIV Put Data (osdp_PIVPUTDATA)

The `osdp_PIVPUTDATA` command (command code `0xA8`) instructs the PD to deliver an ISO 7816-4 `PUT DATA` operation to the credential. The ACU **SHALL** construct the payload as a complete BER-TLV object, including the desired outer tag (`0x7E` for the Discovery Object, `0x7F61` for a BIT Group Template, `0x5C`/`0x53` sequences for other data objects, and so on). The PD **SHALL** forward the payload to the credential using `CLA = 0x00`, `INS = 0xDB`, `P1 = 0x3F`, and `P2 = 0xFF`. If VCI secure messaging is established, the PD **SHALL** select the secure channel automatically and adjust the `CLA` accordingly. Chained APDUs **SHALL** be generated by the PD when the payload exceeds a single-card APDU.

Table 10: PIV Put Data (osdp_PIVPUTDATA)

Size	Name	Meaning	Value
1	CMND	Command identifier.	`0xA8 (tentative)`
2	MpSizeTotal	Total size of the complete PUT DATA payload, least-significant byte first.	`0x0000–0xFFFF`
2	MpOffset	Offset of this fragment within the complete payload, least-significant byte first.	`0x0000–0xFFFF`
2	MpFragmentSize	Number of payload bytes carried in this fragment, least-significant byte first.	`0x0000–0xFFFF`
0-n	Data	BER-TLV payload to be written to the credential. This field already includes the outer object tag selected by the ACU.	`0x00–0xFF`

Responses A successful write is acknowledged with `osdp_ACK`. If the PD does not implement the command, it **SHALL** return `osdp_NAK` function-not-supported. When a credential-side failure occurs, the PD **SHALL** report the failure—either immediately or on a subsequent `osdp_POLL`—using `osdp_PDERROR` and one of the PIV error codes defined in Subsection 2.2: `0x1020` when the function is not supported by the credential, `0x1021` when the PD lacks buffer space to stage the payload, `0x1022` when the credential has insufficient storage, `0x1023` when security conditions are not met, or `0x1027` when the credential is not present or is removed during the operation. The status word is optional.

3.2 PIV Application Select (osdp_PIVSELECT)

The `osdp_PIVSELECT` command (command code `0xB0`, tentative) instructs the PD to select the application identified by the supplied Application Identifier (AID) on the currently active credential. The command provides an OSDP-native selector for ISO/IEC 7816-4 `SELECT by DF name`; it is not a raw APDU tunnel and does not expose APDU header bytes to the ACU. PDs **SHALL** advertise support via Function Code 12 bit `0x04` (Enhanced PIV) before controllers attempt to use this command.

Table 11: PIV Application Select (osdp_PIVSELECT)

Size	Name	Meaning	Value
1	CMND	Command identifier.	`0xB0 (tentative)`
1	Selection Flags	Selection controls. All bits are reserved for future use and **SHALL** be transmitted as zero.	`0x00`
1	AID Length	Length of the Application Identifier.	`0x05-0x10`
5-16	AID	Application Identifier to select by ISO/IEC 7816 DF name.	`0x00-0xFF`

The PD **SHALL** issue `SELECT` with `CLA = 0x00`, `INS = 0xA4`, `P1 = 0x04`, `P2 = 0x00`, `Lc = AID Length`, `Data = AID`, and `Le = 0x00`. The PD **SHALL** request normal FCI return data from the credential and **SHALL** use the active credential interface selected by the current PIV mode or by the PD's card-present workflow. If VCI secure messaging is established and remains valid for the selected application, the PD **SHALL** select the secure channel automatically and adjust the `CLA` accordingly; otherwise changing applications **SHALL** clear VCI-dependent state, cached pairing codes, and any selected-application metadata derived from the prior application.

The *AID Length* field **SHALL** be in the ISO/IEC 7816 application identifier range of 5 to 16 bytes. If the payload is malformed, the AID length is outside that range, or any reserved selection flag is nonzero, the PD **SHALL** report `osdp_PDERROR` `0x0005`. A successful selection is acknowledged with `osdp_ACK`; after success, subsequent PIV commands operate against the selected application until the credential is removed, a new `osdp_PIVSELECT` succeeds, or `osdp_PIVMODE` changes the operating context. The PD **SHALL** update the selected AID reported by `osdp_CARDSTATUSR`; if the credential returns an FCI template containing tag `0x4F`, that value **SHALL** be reported, otherwise the PD **SHALL** report the requested AID.

If the PD does not implement the command, it **SHALL** return `osdp_NAK` function-not-supported. Credential-side failures **SHALL** be reported using `osdp_PDERROR`: `0x1020` when the credential does not support application selection, `0x1023` when security conditions are not satisfied, `0x1024` when the requested AID is not present, or `0x1027` when the credential is not present or is removed during selection. The status word is optional.

3.3 Card Status (osdp_CARDSTATUS)

The osdp_CARDSTATUS command (command code 0xA9, tentative) requests the current credential context from the PD. PDs **SHALL** advertise support via Function Code 12 bit 0x04 (Enhanced PIV) before controllers attempt to use this command. The payload selects the status record that should be returned (currently only the standard profile).

Table 12: Card Status Request (osdp_CARDSTATUS)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xA9 (tentative)
1	Status Type	Requested status record. 0x01 selects the standard PIV card status profile. Values 0x02-0x7F are reserved and shall elicit osdp_PDERROR. Values 0x80-0xFF are reserved for private use.	0x00-0xFF

PDs **SHALL** return osdp_NAK function-not-supported when the command itself is not implemented. If the requested status type is not supported or the payload is malformed, PDs **SHALL** report the condition via osdp_PDERROR (for example 0x0005 incorrect parameter) on a subsequent osdp_POLL.

3.4 Card Status Reply (osdp_CARDSTATUSR)

In response to osdp_CARDSTATUS, the PD reports a ready-to-proceed credential state after completing its card-present workflow (power, application selection, and required reads). PDs **SHALL NOT** send osdp_CARDSTATUSR unsolicited on osdp_POLL. Controllers **SHOULD** continue to use poll-based osdp_RAW replies as the single aggregated "card presented and data acquired" indication; PIN usage policy and VCI connectivity **SHALL NOT** be included in unsolicited poll indications. Detailed readiness (credential type, selected AID, cached data, and readiness flags) **SHALL** be returned only in solicited osdp_CARDSTATUSR. When VCI is established, the reply reports both the secure messaging CVC IIN and the loaded anchor IIN used by the PD. These values are identical for direct CVC verification and differ when an Intermediate CVC is used. The PD **SHALL** set the *VCI Trust Anchor ID* to the identifier assigned in osdp_PIVVCILOADTA; this value **SHALL** correspond to an entry in the osdp_PIVVCITASTATUSR inventory whose IIN matches the *VCI Anchor IIN* field, so that the ACU can correlate reported VCI state with the anchor management record.

Table 13: Card Status Reply (osdp_CARDSTATUSR)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	0x85 (tentative)
1	Detected Credential Type	Current card interface: 0x00 none, 0x01 ISO 7816 contact, 0x02 ISO 14443 contactless, 0x03 other ISO 7816 credential (for example DESFire, Seos), 0x04 low-frequency credential, 0x05 unknown type, 0x06-0x7F reserved, 0x80-0xFF private use.	0x00–0xFF
1	VCI Status	0x00 VCI not established, 0x01 VCI established. Values 0x02-0x7F reserved, 0x80-0xFF private use.	0x00–0xFF
1	VCI Trust Anchor ID	The Trust Anchor ID used by the PD during VCI establishment. The PD **SHALL** set this field to the identifier assigned in osdp_PIV_VCILOADTA; this value **SHALL** correspond to an entry in the osdp_PIV_VCITASTATUSR inventory whose IIN matches the *VCI Anchor IIN* field. Set to 0x00 when VCI is not established.	0x00–0xFF
8	VCI CVC IIN	Issuer Identification Number from the secure messaging CVC tag 0x42. Set to all zero when VCI is not established.	0x00–0xFF
8	VCI Anchor IIN	Issuer Identification Number of the loaded VCI trust anchor used by the PD. Set to all zero when VCI is not established or no anchor was used.	0x00–0xFF
2	PIN Usage Policy	PIN Usage Policy bytes (tag 0x5F2F) returned verbatim when available. A value of 0xFFFF indicates the policy is not present or has not yet been read.	0x0000–0xFFFF
1	Selected AID Length	Length (0-16) of the selected AID. 0x00 indicates no application is currently selected.	0x00–0x10

Size	Name	Meaning	Value
0-16	Selected AID	Application Identifier read from the credential (for example from the FCI template). Present only when the length is non-zero.	0x00–0xFF

A PD that cannot provide the requested status **SHALL** return osdp_NAK when the command is unsupported. For credential-side failures encountered while preparing status, the PD **MAY** omit osdp_CARDSTATUSR data and, if available, report details via osdp_PDERROR on a subsequent osdp_POLL. Successful replies include the entire selected AID (when available) and the fixed two-byte PIN Usage Policy field after the PD has completed its preparation. When PIN Usage Policy bytes from the credential's Discovery Object (tag 0x5F2F) are available, the PD **SHALL** report those two bytes verbatim. When no PIV PIN Usage Policy is present or the policy has not yet been read, the PD **SHALL** report 0xFFFF. The VCI fields describe stable readiness (for example whether a secure channel is established) and the IIN evidence the ACU needs for validation; transient negotiation state **SHALL NOT** be reported. If the credential is removed before preparation completes, the PD **MAY** report the condition via osdp_PDERROR (for example 0x1027) instead of osdp_CARDSTATUSR. osdp_CARDSTATUSR is sent only in response to a matching osdp_CARDSTATUS request.

3.5 PIV Secure PIN Entry (osdp_PIVSPE)

The osdp_PIVSPE command (command code 0xAA, tentative) controls Secure PIN Entry for the Enhanced PIV profile. A PD **SHALL** advertise both Function Code 12 bit 0x04 (Enhanced PIV) and Function Code 15 Secure PIN Entry support before an ACU uses this command. The OSDP 2.2 osdp_XWR Mode-01 SPE command and osdp_PR01SPER reply remain unchanged.

The ACU is responsible for conveying a clear indication that PIN entry is in progress. While the PD is acquiring a PIN, it **SHALL** suppress osdp_KEYPAD reports so that no PIN digit is sent to the ACU.

Table 14: PIV Secure PIN Entry (osdp_PIVSPE)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xAA (tentative)
1	SPE Mode	Secure PIN Entry operating mode. See Table 15.	0x00–0xFF
1	Start Timeout	Seconds allowed to begin key entry; 0x00 selects a finite PD default.	0x00–0xFF
1	Inter-key Timeout	Seconds allowed between keystrokes; 0x00 selects a finite PD default.	0x00–0xFF
1	Flags	Behaviour modifiers. See Table 16.	0x00–0xFF
2	Cache Validity	PIN cache validity in seconds (LSB first). Mode 0x04 requires 0x0001-0xFFFF; other modes transmit 0x0000.	0x0000–0xFFFF

Table 15: SPE Mode Values

`Value`	Description
`0x00`	Report the persistent SPE status without initiating PIN entry.
`0x01`	Acquire a PIN and verify it with the credential currently present.
`0x02`	Execute VCI PIN verification using a cached PIN.
`0x03`	Execute normal PIN verification using a cached PIN.
`0x04`	Acquire and cache a PIN.
`0x05`	Abort active PIN entry and clear the cached PIN.
`0x06`	Report attempts remaining without initiating PIN verification.
`0x07-0x7F`	Reserved for future use.
`0x80-0xFF`	Private use.

Table 16: SPE Flag Values

`Mask`	Description
`0x02`	Retain a cached PIN across credential removal until its validity expires or another clearing event occurs. Valid only with Mode `0x04`.
`0x08`	Auto Execute When Credential Is Presented. Valid only with verification Modes `0x01`-`0x03`.
`0x01, 0x04, 0x10-0x80`	Reserved and **SHALL** be transmitted as zero.

Modes are enumerated values, not a bit mask. Mode `0x02` requires VCI; Mode `0x03` uses the normal credential interface selected by the active PIV mode. Mode `0x04` requires a *Cache Validity* from `0x0001` through `0xFFFF`. All other modes **SHALL** transmit `0x0000` in that field. The PD **SHALL NOT** transmit an entered or cached PIN to the ACU.

A zero *Start Timeout* or *Inter-key Timeout* selects a finite, PD-defined default; neither value permits an infinite timer. The remaining timeout values specify seconds. Fields that do not apply to a selected mode **SHALL** be transmitted as zero.

Only one `osdp_PIVSPE` operation may be outstanding. Modes `0x01`-`0x04` each initiate one operation. The PD returns `osdp_ACK` when it accepts the request, then returns one `osdp_PIVSPER` in response to a later `osdp_POLL`. While such an operation is pending, the ACU **SHALL** send only `osdp_POLL`, Mode `0x05`, or `osdp_ABORT`. Returning `osdp_PIVSPER` ends the operation unconditionally, including after a PIN mismatch.

Mode `0x00` returns `osdp_PIVSPER` directly. It reports only persistent PIN state; it does not report transient active or waiting states. An ACU recovering from an interrupted session first uses Mode `0x05` or `osdp_ABORT`, then queries Mode `0x00`. Mode `0x06` is accepted with `osdp_ACK`; the PD reads the present credential without performing PIN verification and returns the count in a subsequent osdp_PIVSPER. This query **SHALL NOT** decrement the credential retry counter. Auto Execute does not make the count available without a present credential.

Mode 0x05 is idempotent. It aborts active or partial PIN entry, clears the cached PIN, and completes with osdp_ACK; the PD **SHALL NOT** send a later osdp_PIVSPER for the canceled operation. osdp_ABORT cancels active or partial PIN entry but preserves a cache that was completed before the active operation.

A cached PIN is used, not consumed. It remains available until its validity expires or it is cleared. The PD **SHALL** clear it on Mode 0x05, PD reset, loss of the OSDP secure channel or applicable security context, or a change to the PIV mode or selected application. Credential removal also clears it unless flag 0x02 was supplied with Mode 0x04. PIN-verified status is bound to the current credential, selected application, interface, and credential security context. The PD **SHALL** clear that status on credential removal or reset, application selection or reselection, interface change, PIV mode change, or VCI teardown, and **SHALL NOT** infer it only from an earlier successful reply.

Flag 0x08 arms the verification selected by Mode 0x01, 0x02, or 0x03 to execute when a credential is presented. It **SHALL** be zero for Modes 0x00, 0x04, 0x05, and 0x06. A cached auto-execute workflow first acquires the PIN with Mode 0x04, then selects normal or VCI verification with Mode 0x03 or 0x02 and flag 0x08.

The PD **SHALL** reject an unsupported command with osdp_NAK 0x03, an incorrect command length with osdp_NAK 0x02, and an invalid mode, reserved bit, or field combination with osdp_NAK 0x09. Accepted credential-side outcomes are reported through osdp_PIVSPER, not osdp_PDERROR.

3.6 PIV Secure PIN Entry Reply (osdp_PIVSPER)

The osdp_PIVSPER reply (reply code 0x86, tentative) contains one mode-dependent *Result* byte. Removing the former third byte affects only this Enhanced PIV reply; the OSDP 2.2 osdp_PR01SPER *Tries* byte is unchanged.

Table 17: PIV Secure PIN Entry Reply (osdp_PIVSPER)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	0x86 (tentative)
1	Result	Mode-dependent SPE status, attempts count, or operation result.	0x00–0xFF

For Mode 0x00, *Result* is the status bitmap in Table 18. For Mode 0x06, it is the attempts value in Table 19. For Modes 0x01-0x04, it is the operation outcome in Table 20.

Table 18: PIVSPE Status Result (Mode `0x00`)

`Value`	Description
`0x00`	No cached PIN is present and the current credential is not PIN verified.
`0x01`	A cached PIN is present.
`0x02`	The current credential and selected application are PIN verified.
`0x03`	A cached PIN is present and the current credential and selected application are PIN verified.
`0x04-0x7F`	Reserved for future use.
`0x80-0xFF`	Private use.

Table 19: PIVSPE Attempts Result (Mode `0x06`)

`Value`	Description
`0x00`	The credential PIN is blocked.
`0x01-0xFE`	Number of PIN verification attempts remaining.
`0xFF`	Attempts remaining are unavailable, including when no credential is present.

Table 20: PIVSPE Result Codes

`Value`	Description
`0x00`	Requested operation completed.
`0x01`	Timeout – PIN entry did not begin within the allotted time.
`0x02`	Timeout – inter-digit interval exceeded.
`0x03`	PIN mismatch.
`0x04`	Credential removed during operation.
`0x05`	Credential PIN blocked.
`0x06`	Wrong interface for the requested mode.
`0x07`	Incorrect parameters.
`0x08`	Security condition not met.
`0x09`	VCI not established for the requested mode.
`0x0A`	No cached PIN is available.
`0x0B-0x7F`	Reserved for future use.
`0x80-0xFF`	Private use.

3.7 PIV Biometric Match (osdp_PIVBIOMATCH)

The osdp_PIVBIOMATCH command (command code 0xAB, tentative) instructs the PD to acquire a live sample and compare it to the specified biometric template from the active application. Depending on the *Match Mode*, the comparison occurs either on the credential (on-card match) or within the PD (off-card match).

Table 21: PIV Biometric Match (osdp_PIVBIOMATCH)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xAB (tentative)
1	Biometric Type	Selects the template type to compare. See Table 22.	0x00-0xFF
1	Match Mode	0x00 off-card match; 0x01 on-card match.	0x00-0x01

Table 22: Biometric Type Values

Value	Description
0x01	Cardholder fingerprint template.
0x02	Cardholder facial image.
0x04	Cardholder iris template.

Controllers set *Match Mode* to select off-card or on-card comparison. Off-card flows require the PD to retrieve the appropriate template and, where applicable (e.g., TWIC contactless), to have a valid privacy key loaded via osdp_PIVLOADTPK. On-card flows require the credential to support OCC. If the PD does not implement the command, it **SHALL** return osdp_NAK function-not-supported. Credential-side match outcomes (for example PIN verification required or credential removal during operation) **SHALL** be reported in the subsequent osdp_PIVBIOMATCHR result code per Table 24. Conditions that prevent the command from being attempted (for example unsupported credential function, missing VCI when required, or invalid command parameters) **MAY** be reported via osdp_PDERROR on a subsequent osdp_POLL.

3.8 PIV Biometric Match Reply (osdp_PIVBIOMATCHR)

After osdp_PIVBIOMATCH completes, the PD reports the result as a poll response using reply code 0x87 (tentative).

Table 23: PIV Biometric Match Reply (osdp_PIVBIOMATCHR)

Size	Name	Meaning	`Value`
1	CMND	Reply identifier.	`0x87 (tentative)`
1	Result	Outcome of the biometric match. See Table 24.	`0x00-0xFF`
0 or 32	Hash	32-byte SHA-256 hash of the biometric template tag `0xBC` data field. The PD **SHALL** include this field only when *Result* is `0x00` (off-card match success). The PD **SHALL** omit this field for on-card match success (`0x09`) and all failure result codes.	`0x00-0xFF`

When *Result* is `0x00` (off-card match success), the PD **SHALL** include the 32-byte SHA-256 hash of the biometric template tag `0xBC` data field in the reply. For on-card match success (*Result* `0x09`) and for all failure result codes, the PD **SHALL** omit the hash field; ACUs **SHALL NOT** rely on a hash received alongside any result other than `0x00`.

Off-card biometric comparison demonstrates only that the live sample matched template data presented to the PD. Because off-card biometric objects can be copied or duplicated, an off-card biometric match does not by itself prove authenticity of the credential. On-card biometric comparison is a credential-controlled assertion; the biometric object may be attacker controlled and the PD does not receive the template data being matched. To obtain cryptographic proof associated with an on-card biometric success, the ACU **SHALL** perform an additional authentication using an appropriately verified credential key after the successful on-card comparison.

Table 24: PIVBIOMATCH Result Codes

`Value`	Description
`0x00`	Biometric match success (off-card match).
`0x01`	Biometric match timeout (no entry started).
`0x02`	Biometric match failed (no match).
`0x03`	PIN verification required.
`0x04`	Credential removed during operation.
`0x05`	PIN verification failed.
`0x06`	PIN verification timeout.
`0x07`	Reserved for future use.
`0x08`	Reserved (PD/credential errors are reported via `osdp_PDERROR`).
`0x09`	On-card match success.

Value	Description
0x0A-0x7F	Reserved for future use.
0x80-0xFF	Private use.

3.9 TWIC Privacy Key Load (osdp_PIVLOADTPK)

The osdp_PIVLOADTPK command (command code 0xAC, tentative) instructs the PD to cache a TWIC Privacy Key (TPK) for use with subsequent biometric operations against a TWIC credential. Function Code 14 is the biometric-support compliance byte; before using this command, controllers **SHOULD** verify that Function Code 14 advertises the OSDP 3.0 biometric command set and the relevant biometric modality bits for the intended TWIC operation.

Table 25: TWIC Privacy Key Load (osdp_PIVLOADTPK)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xAC (tentative)
1	Cache Timeout	Number of seconds to retain the cached TPK. 0x00 clears any cached key immediately; 0x01-0xFF permit caching for up to 255 seconds.	0x00-0xFF
1	Control Flags	Bit 0 set requests automatic clearing when the credential is removed; all other bits **SHALL** be zero.	0x00-0xFF
32	TPK	Trusted Processing Key bytes to cache for the next TWIC biometric operation.	0x00-0xFF

Controllers set *Cache Timeout* to define how long (in seconds) the PD may retain the key. A value of 0x00 clears any cached TPK immediately. Values 0x01-0xFF allow caching for up to 255 seconds. Bit 0 of *Control Flags* directs the PD to clear the cached TPK when the credential is removed; all other bits are reserved and **SHALL** be transmitted as zero. If the PD does not support TWIC TPK caching it **SHALL** return osdp_NAK function-not-supported. Invalid payloads (for example reserved bits set) **SHALL** elicit osdp_PDERROR 0x0005.

3.10 PIV VCI Pairing Code Transmit (osdp_PIVXMITPAIRING)

The osdp_PIVXMITPAIRING command (command code 0xAD, tentative) delivers an eight-digit pairing code to the PD for use during Virtual Contact Interface (VCI) establishment. The pairing code is transmitted from the ACU to the PD only over OSDP Secure Channel 2.0, and is used by the PD to establish VCI secure messaging with the credential. Controllers **SHALL NOT** issue this command unless a PIV application is currently selected and VCI mode is active.

Table 26: PIV VCI Pairing Code Transmit (osdp_PIVXMITPAIRING)

Size	Name	Meaning	Value
1	CMND	Command identifier.	`0xAD (tentative)`
8	Pairing Code	Eight ASCII digits (`'0'`-`'9'`) used by the PD during VCI establishment for the currently selected PIV application.	`0x30–0x39`

The pairing code is encoded as eight ASCII digits (`'0'`-`'9'`). If no PIV application is selected, the PD **SHALL** return `osdp_PDERROR` `0x0005`. Otherwise, the PD **SHALL** cache the pairing code only as an input to VCI establishment for the currently selected PIV application and use it when required by the credential's Discovery Object PIN Usage Policy (for example when bit 4 is set and bit 3 is clear in the first byte). The pairing code is not evidence that VCI secure messaging is already established, and `osdp_PIVXMITPAIRING` is not transported over VCI secure messaging. When VCI secure messaging is already active or the credential does not require pairing to establish VCI, the PD **MAY** ignore the supplied code after acknowledging receipt. PDs **SHALL** clear any stored pairing code when VCI is established, when VCI is torn down, or when the active PIV mode changes.

VCI security preconditions VCI trust-anchor load/status, VCI pairing-code transmit, VCI establishment, and PIV operations that depend on VCI **SHALL** be accepted only when OSDP Secure Channel 2.0 is active between the ACU and PD. This OSDP Secure Channel protects ACU-to-PD management traffic, including `osdp_PIVXMITPAIRING`; VCI secure messaging is the separate PD-to-credential secure channel established by the PD. A PD **SHALL** reject VCI-related functionality when operating with factory/default SCBK or other factory keys. Security-precondition failures **SHALL** be reported using `osdp_PDERROR` `0x0006` where the command supports `osdp_PDERROR`; otherwise the command's existing failure reporting path applies.

3.11 Set PIV Mode (osdp_PIVMODE)

The `osdp_PIVMODE` command (command code `0xAE`, tentative) defines the operating context for PIV, TWIC, and related credential transactions. OSDP 3.0 replaces the OSDP 2.2 "Application + AID" fields with a configuration payload that includes a required Global Interface Mask byte, PIV Auto challenge parameters when autonomous operation is enabled, and a TLV list of configuration descriptors so controllers can specify multiple application profiles, preferred interfaces, and optional Application Identifiers. PDs **SHALL** advertise support via Function Code 12 bit `0x04` (Enhanced PIV).

Table 27: Set PIV Mode (osdp_PIVMODE)

Size	Name	Meaning	Value
1	CMND	Command identifier.	`0xAE (tentative)`
1	PIV Mode Flags	Global behaviour flags (Section 3.11).	`0x00–0xFF`
2	MpSizeTotal	Total size of the complete configuration payload, least-significant byte first.	`0x0000–0xFFFF`
2	MpOffset	Offset of this fragment within the configuration payload, least-significant byte first.	`0x0000–0xFFFF`
2	MpFragmentSize	Length of the fragment payload, least-significant byte first.	`0x0000–0xFFFF`
1	Config Count	Number of configuration entries (1-32).	`0x01–0x20`
1	Global Interface Mask	Bitmask limiting interfaces available to all configurations (bit 0 contact, bit 1 contactless, bit 3 Bluetooth; bit 2 reserved and set to zero).	`0x00–0x0B`
4	Series Counter	Conditional field present only when PIV Auto is enabled. Least-significant byte first.	`0x00000000 –0xFFFFFFFF`
4	Sequence Counter	Conditional field present only when PIV Auto is enabled. Least-significant byte first.	`0x00000000 –0xFFFFFFFF`
32	Supplemental Entropy	Conditional fixed-width field present only when PIV Auto is enabled. All zeroes are permitted but not recommended.	`32 octets`
0-n	TLV Data	Sequence of Configuration Entry TLVs (Tag `0x01`) that follow the Global Interface Mask, or follow the conditional PIV Auto fields when PIV Auto is enabled.	–

The *PIV Mode Flags* byte enables global behaviours such as autonomous operation and VCI usage (Table 28). Bits not listed remain reserved and **SHALL** be transmitted as zero.

Table 28: PIV Mode Flags

`Mask`	Description
`0x01`	Enable autonomous PIV (PIV Auto). The PD interrogates credentials, establishes VCI when possible, derives a KMAC256 challenge from the configured counters and entropy, and reports the signed response using `osdp_PIVSTATUS`.
`0x02`	Permit VCI usage when available.
`0x04`	Enforce FICAM strict mode (PIV applications only).

Global and per-entry interface masks use the bit positions in Table 29; the Global Interface Mask bounds the interfaces a PD may use when processing all configuration entries.

Table 29: PIV Mode Interface Masks

`Mask`	Description
`0x01`	Permit ISO 7816 contact credential interface use.
`0x02`	Permit ISO 14443 contactless credential interface use.
`0x04`	Reserved and **SHALL** be transmitted as zero.
`0x08`	Permit Bluetooth credential interface use.
`0x10–0x80`	Reserved and **SHALL** be transmitted as zero.

After the flags, the payload uses standard multi-part framing fields and a configuration count (1-32). The count is followed by the required Global Interface Mask byte. If and only if PIV Auto is enabled in *PIV Mode Flags*, the Global Interface Mask is followed by the Series Counter, Sequence Counter, and fixed 32-byte Supplemental Entropy field before the TLV list. When PIV Auto is not enabled, those three conditional fields are absent and the TLV list begins immediately after the Global Interface Mask. Each configuration appears as a TLV:

- **Global Interface Mask (required)** — One-byte mask (no tag) that limits the interfaces the PD may use for all configurations (see Table 29). A value of `0x00` indicates no global restriction; reserved bits **SHALL** be zero.
- **Configuration Entry (Tag `0x01`)** — Contains the data-model identifier, a per-entry interface mask, the AID length, and the optional AID:
  - *Data Model* (`0x01` PIV, `0x02` TWIC, `0x03` CAC, values `0x04`-`0x7F` reserved, `0x80`-`0xFF` private use). CAC entries use the same interface and AID encoding as PIV and TWIC entries.
  - *Interface Mask* — Uses the same bit positions as the global mask (Table 29) and **SHALL NOT** assert interfaces excluded by the Global Interface Mask. A value of `0x00` means "any interface permitted by the global mask."
  - *AID Length* (0-16) and *AID* bytes when provided.

PDs process configuration entries in order, applying the global mask (when present) before honouring each entry's mask. Invalid masks, unknown data-model identifiers, or malformed TLVs **SHALL** result in `osdp_PDERROR` `0x0005`. Controllers **SHALL** send the command over a secure channel where required; the PD **SHALL** return `osdp_PDERROR` `0x0006` if the necessary security conditions are not satisfied. Multi-part framing errors result in `osdp_PDERROR` `0x0007`. If the PD cannot cache the requested configuration set, it **SHALL** respond with `osdp_PDERROR` `0x1021`.

When PIV Auto is enabled, PDs interrogate the credential on presentation, establish VCI automatically when possible, derive the PIV Auto challenge, issue the PIV GENERAL AUTHENTICATE command, and report the signed response without additional ACU prompts. If a pairing code is required, the PD signals the state change via a solicited `osdp_CARDSTATUSR` (not via unsolicited poll); the ACU then provides the code using `osdp_PIVXMITPAIRING` over OSDP Secure Channel 2.0, and the PD uses the code to complete VCI establishment.

PIV Auto configuration **SHALL** be accepted only when OSDP Secure Channel 2.0 is active and the channel is not using a factory/default SCBK or other factory key material. A PD **SHALL** reject PIV Auto configuration or execution with `osdp_PDERROR` `0x0006` when these preconditions are not met.

When bit `0x01` of *PIV Mode Flags* enables PIV Auto, the configuration **SHALL** specify a *Series Counter*, a *Sequence Counter*, and a 32-byte *Supplemental Entropy* field. The supplemental entropy field is fixed width and **SHALL** always be present; all zeroes are permitted for compatibility but are not recommended. The ACU **SHALL** ensure that a series counter value does not repeat for the same PD and that no sequence counter value is issued more than once within a series. One conforming implementation strategy is to maintain a monotonic series counter that increments at every ACU boot, similar to the SNMP engine boots counter, and a monotonic sequence counter for challenges issued within that boot series.

The PIV Auto challenge **SHALL** be derived from the active Secure Channel 2.0 session key material using two KMAC256 operations. First, the PD derives a 32-byte PIV Auto KDK with key material formed by concatenating `S-ENC`, `S-MAC1`, and `S-MAC2`; empty input data; output length 32 bytes; and customization string `OSDP-PIV-AUTO-KDK-v1`. Second, the PD derives the challenge digest using the PIV Auto KDK as the KMAC key, the fixed-width template as input data, output length `L`, and customization string `OSDP-PIV-AUTO-CHALLENGE-v1`. The fixed-width template **SHALL** contain the series counter, sequence counter, credential FASC-N, credential UUID, challenged PIV key reference, PIV algorithm identifier, and the 32-byte supplemental entropy field. A successful PIV Auto transaction **SHALL** include the credential UUID; if the UUID cannot be read, the PD **SHALL** report PIV Auto failure instead of deriving a success challenge with an all-zero UUID. The output length `L` is 32 bytes for SHA-256 profiles and 48 bytes for SHA-384 profiles. PIV Auto challenge derivation **SHALL NOT** use static factory keys.

3.12 PIV Auto Status Reply (osdp_PIVSTATUS)

After a PIV Auto card challenge completes, the PD reports the result as a poll response using reply code `0x89` (tentative). The first `osdp_POLL` response after completion **SHALL** contain `osdp_PIVSTATUS`; subsequent fragments, when required, **SHALL** use the same reply and advance *MpOffset* until the complete status payload is delivered. This is an autonomous completion reply; there is no corresponding `osdp_PIVSTATUS` request command.

Table 30: PIV Auto Status Reply (osdp_PIVSTATUS)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	`0x89 (tentative)`
2	MpSizeTotal	Total size of the complete PIV Auto status payload, least-significant byte first.	`0x0000-0xFFFF`
2	MpOffset	Offset of this fragment within the complete status payload, least-significant byte first.	`0x0000-0xFFFF`
2	MpFragmentSize	Length of the status payload bytes in this fragment, least-significant byte first.	`0x0000-0xFFFF`
1	Result	PIV Auto result for the challenged credential. See Table 31.	`0x00-0xFF`
1	Detail	Additional result detail. Reserved as `0x00` when Result is `0x00`; may carry SW1|SW2 low byte when Result is `0x06`.	`0x00-0xFF`
4	Series Counter	Series counter value used to derive the challenge, least-significant byte first.	`0x00000000 -0xFFFFFFFF`
4	Sequence Counter	Sequence counter value used to derive the challenge, least-significant byte first.	`0x00000000 -0xFFFFFFFF`
25	FASC-N	FASC-N read from the credential.	`200-bit FASC-N`
16	UUID	Credential UUID read from the credential. Success replies **SHALL** carry a nonzero UUID; failure replies may transmit all zeroes when the UUID is unavailable.	`16 octets`
1	Key Reference	PIV key reference challenged by the PD.	`For example 0x9E`
1	Algorithm Identifier	PIV GENERAL AUTHENTICATE algorithm identifier.	`For example 0x07`
2	Signed Response Length	Total length of the signed response, least-significant byte first.	`0x0000-0xFFFF`
0-n	Signed Response Fragment	Contiguous bytes of the signed response carried in this fragment.	`Variable`

Result and detail codes follow Table 31. When *Result* is `0x00`, *Detail* is reserved and **SHALL** be `0x00`. When *Result* is not `0x00`, the PD **SHALL** set *Signed Response Length* to `0x0000` unless a credential response was received and is useful for ACU diagnostics.

Table 31: PIV Auto Status Result Codes

`Value`	Description
`0x00`	Success. *Detail* is `0x00`, UUID is nonzero, and Signed Response carries the complete returned Dynamic Authentication Template.
`0x01`	Credential not present or removed before PIV Auto completed.
`0x02`	VCI establishment or required VCI validation failed.
`0x03`	Required credential data object, FASC-N, UUID, key, or certificate evidence is unavailable.
`0x04`	Requested PIV key reference or algorithm is not supported for PIV Auto.
`0x05`	Credential security status is not satisfied, including missing PIN or pairing preconditions.
`0x06`	Credential GENERAL AUTHENTICATE operation failed; *Detail* may carry SW1|SW2 when available.
`0x07`	Signed response verification failed.
`0x08`	Timeout or PD internal failure before a definitive credential result was available.
`0x09–0x7F`	Reserved for future standard result codes.
`0x80–0xFF`	Private use.

The reply reports the exact series and sequence counters used for the challenge, the FASC-N and UUID read from the credential, and the signed response returned by the credential. The ACU uses these fields to verify counter uniqueness, bind the response to the credential identity, and validate the signature and certificate path using ACU trust policy. A success result **SHALL** include a nonzero credential UUID. The PD **SHALL** preserve the credential's signed response bytes as returned by the PIV operation, excluding transport status words such as SW1/SW2.

If the complete signed response does not fit in one OSDP response, the PD **SHALL** fragment `osdp_PIVSTATUS` using the OSDP 2.2 multi-part convention. *MpSizeTotal* is the total status payload size, *MpOffset* is the zero-based offset of the fragment, and *MpFragmentSize* is the number of payload bytes carried in that fragment. The fixed fields appear at offset zero; later fragments carry the next contiguous bytes of the signed response.

3.13 VCI Trust Anchor Load (osdp_PIVVCILOADTA)

The `osdp_PIVVCILOADTA` command (command code `0xAF`, tentative) loads, replaces, or deletes a Virtual Contact Interface (VCI) trust anchor. The ACU **SHALL** validate and authorize each trust anchor under its deployment certificate policy before loading it. Trust anchors must be cached before a PD can verify card certificates during VCI establishment.

Table 32: **VCI Trust Anchor Load (osdp_PIVVCILOADTA)**

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xAF (tentative)
1	Operation	0x00 load/replace anchor; 0x01 delete anchor. Anchor inventory is requested with osdp_PIV_VCITASTATUS.	0x00–0x01
1	Trust Anchor ID	Controller-assigned identifier for the anchor (1-128).	0x00–0xFF
8	Issuer Identifier Number	The leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate. This value **SHALL** match the IIN field in the VCI trust anchor record (tag 0x42) and is used as the anchor selector during VCI establishment.	0x00–0xFF
2	Anchor Length	Total size of the variable-length VCI trust anchor record, least-significant byte first. Set to 0x0000 when deleting.	0x0000–0x0240
2	MpSizeTotal	Total size of the complete anchor data, least-significant byte first. For load operations, this value equals AnchorLength. Set to 0x0000 when deleting.	0x0000–0x0240
2	MpOffset	Offset of this fragment within the anchor data, least-significant byte first. Set to 0x0000 when deleting.	0x0000–0x023F
2	MpFragmentSize	Length of the data block in this fragment, least-significant byte first. MpOffset + MpFragmentSize **SHALL NOT** exceed AnchorLength. Set to 0x0000 when deleting.	0x0000–0x0240
0-n	Anchor Data	Fragment bytes of the OSDP-defined VCI trust anchor record. Omitted for delete operations.	0x00–0xFF

The metadata block (operation, identifier, IIN, anchor length) appears only in the first fragment where MpOffset = 0; subsequent fragments carry additional anchor bytes. Controllers set *Operation* to 0x00 to load or replace the anchor identified by *Trust Anchor ID*. Set *Operation* to 0x01 with *Anchor Length* 0x0000 to delete the anchor. Inventory is requested with osdp_PIVVCITASTATUS, not with osdp_PIVVCILOADTA. The *Issuer Identifier Number (IIN)* is an OSDP-defined anchor selector: it **SHALL** be the leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate associated with this anchor. This value **SHALL** match the IIN field (tag 0x42) in the VCI trust anchor record.

The *Anchor Length* declares the exact total size of the variable-length VCI trust anchor record. For load operations, MpSizeTotal **SHALL** equal *Anchor Length*, the sum of all accepted fragment lengths **SHALL** equal *Anchor Length*, and no fragment **SHALL** extend beyond *Anchor Length*. A conforming VCI trust anchor record **SHALL NOT** exceed 576 bytes (0x0240). The PD **SHALL** accept any valid record length from the minimum BER-TLV encoding through 0x0240, subject to supported-key and storage-capacity checks.

PD storage accounting for VCI trust anchors is variable-length. A PD **SHALL NOT** treat every anchor as consuming the maximum 0x0240 bytes unless that is the PD's documented physical allocation model. When reporting capacity, loading, replacing, or deleting anchors, the PD **SHALL** account for stored anchors by their actual *Anchor Length* values and **SHALL** use the same accounting rule for both osdp_PIVVCITASTATUSR *Available Bytes* and osdp_PIVVCILOADTA storage acceptance. A replacement operation **SHALL** be evaluated after releasing the storage occupied by the existing anchor with the same *Trust Anchor ID*; if the replacement cannot be completed, the prior anchor **SHALL** remain available. A delete operation **SHALL** release the storage associated with the deleted anchor.

Table 33: VCI Trust Anchor Record Format

Tag	Length	Name	Value
0x7F50	Variable	VCI Trust Anchor Record	OSDP-defined record used to verify a secure messaging CVC or an Intermediate CVC.
0x80	1	Record Profile Identifier	0x01.
0x42	8	Issuer Identifier Number	Anchor selector. This value **SHALL** match the IIN in the osdp_PIV_VCILOADTA metadata.
0x7F49	1-550	SubjectPublicKeyInfo	DER-encoded X.509 SubjectPublicKeyInfo for the public key trusted by the ACU. Supported public-key algorithms are RSA public keys up to 4096 bits and EC public keys for P-256 or P-384.

The VCI trust anchor record is intentionally not a NIST SP 800-73 Intermediate CVC and does not carry X.509 path-validation state. The ACU is responsible for selecting, loading, unloading, and validating the VCI anchors it permits a PD to use before the PD relies on them. The PD uses the record only as a local public-key verifier selected by IIN. The maximum record size is derived from the largest supported key case: a 550-byte RSA-4096 SubjectPublicKeyInfo plus the fixed fields above and BER-TLV length overhead. The resulting worst case is 573 bytes; this specification rounds the maximum accepted record size to 576 bytes.

Use during VCI establishment When the PD receives the secure messaging CVC from the credential, it **SHALL** first extract the Issuer Identification Number from tag 0x42. If a loaded VCI trust anchor has a matching IIN, the PD **SHALL** use that anchor's SubjectPublicKeyInfo to verify the secure messaging CVC signature. If verification succeeds, the PD may continue VCI establishment using the public key in the secure messaging CVC.

If no loaded VCI trust anchor matches the secure messaging CVC IIN, the PD **SHALL** retrieve the Secure Messaging Certificate Signer data object (PIV data object `0x1017`, tag `5FC122`) from the credential and inspect it for one Intermediate CVC (tag `0x7F21`). If the object is absent or no Intermediate CVC is present, validation fails. If an Intermediate CVC is present, the PD **SHALL** verify all of the following before continuing:

- The secure messaging CVC IIN (tag `0x42`) equals the Subject Identifier (tag `0x5F20`) in the Intermediate CVC.
- The Intermediate CVC Role Identifier (tag `0x5F4C`) is `0x12`.
- A loaded VCI trust anchor exists whose IIN equals the Intermediate CVC IIN (tag `0x42`).
- The Intermediate CVC signature verifies using the matching VCI trust anchor public key.
- The secure messaging CVC signature verifies using the public key carried in the Intermediate CVC.

The PD validates only the CVC chain needed for VCI establishment: either the loaded trust anchor directly verifies the secure messaging CVC, or the loaded trust anchor verifies the Intermediate CVC that then verifies the secure messaging CVC. The PD **SHALL NOT** establish VCI unless this CVC validation succeeds. The PD **SHALL NOT** perform full X.509 path validation, validity-period evaluation, certificate-policy evaluation, or revocation processing for VCI trust anchors; those checks are completed by the ACU before loading anchors and again as an acceptance policy after VCI state is reported.

ACU post-VCI validation After receiving `osdp_CARDSTATUSR` with VCI established, the ACU **SHALL** validate the content-signing certificate corresponding to the reported *VCI Anchor IIN* before treating the VCI session as accepted for the deployment. This validation is an additional ACU acceptance check and does not replace the PD's required CVC validation during VCI establishment. The ACU **SHALL** perform all of the following:

- Obtain the content-signing X.509 certificate. If the ACU does not already possess a cached copy whose subjectKeyIdentifier leftmost-eight-bytes equals the *VCI Anchor IIN*, the ACU **SHALL** use `osdp_PIVDATA` to retrieve the Secure Messaging Certificate Signer data object (PIV data object `0x1017`, tag `5FC122`) and extract the certificate from inner tag `0x70`.
- Verify that the leftmost eight bytes of the certificate's subjectKeyIdentifier equal the *VCI Anchor IIN* reported in `osdp_CARDSTATUSR`.
- Verify that the certificate's SubjectPublicKeyInfo matches the SubjectPublicKeyInfo stored in the loaded VCI trust anchor record for that IIN.
- Verify that the certificate includes the PIV content-signing extended key usage OID (`2.16.840.1.101.3.6.7`) or the PIV-I equivalent (`2.16.840.1.101.3.8.7`).
- Verify that the current time falls within the certificate's validity period (`notBefore` $\leq$ now $\leq$ `notAfter`). If the certificate has expired or is not yet valid, the ACU **SHALL** reject the VCI session and **SHALL** unload the corresponding trust anchor.

- Perform X.509 path validation and revocation checking in accordance with the deployment's certificate validation policy.

Table 34: VCI Trust Anchor Examples from NIST Vectors

`Vector`	Content Signer	`Anchor IIN`	SPKI	Record
`card_2`, `card_5`	Test PIV Content Signer 4 (EC P-384, direct)	`B6103792270FBD08`	120 bytes	140 bytes
`card_3`	Test PIV Content Signer 1 (RSA-2048, intermediate)	`9BC5E7735F14EC7F`	294 bytes	317 bytes
`card_4`	Test PIV Content Signer 3 (EC P-256, direct)	`1905976BDDCA823D`	91 bytes	110 bytes
`card_16`	Test PIV-I Content Signer 1 (RSA-2048, intermediate)	`4CC9C7B3A4F1C83E`	294 bytes	317 bytes

In the `card_16` intermediate example, the secure messaging CVC IIN is `6F8FF48F42B3E22E`; the Intermediate CVC Subject Identifier (tag `0x5F20`) is also `6F8FF48F42B3E22E`; and the Intermediate CVC IIN (tag `0x42`) is `4CC9C7B3A4F1C83E`, matching the loaded RSA trust anchor.

PDs **SHALL** reconstruct the anchor from the fragments before storing it. The PD **SHALL** verify that the reconstructed byte string is exactly *Anchor Length* bytes and is a single `0x7F50` record before committing it to storage. A successful operation returns `osdp_ACK`. Failures return:

- `osdp_NAK` function-not-supported when the PD does not support trust-anchor caching.
- `osdp_PDERROR 0x0005` for malformed metadata (unknown operation, reserved bits, invalid lengths, `MpSizeTotal` not equal to *Anchor Length*, or a fragment range outside *Anchor Length*).
- `osdp_PDERROR 0x0007` when multi-part sequencing fails.
- `osdp_PDERROR 0x1021` when the PD lacks storage space for the anchor.

PDs are not required to maintain a real-time clock. PDs may retain either the supplied VCI trust anchor record or an equivalent internal representation so long as the public key and supplied IIN remain available for the verification flow above.

Trust anchors may be stored in volatile or non-volatile memory at the PD's discretion. Controllers should not assume retention beyond the device's documented behaviour.

3.14 VCI Trust Anchor Status (osdp_PIVVCITASTATUS)

The `osdp_PIVVCITASTATUS` command (command code `0xA6`, tentative) enumerates cached trust anchors and reports remaining storage capacity. The request carries no payload. The reply uses the standard multi-part fields (`TOTAL`, `OFFSET`, `DATA_LEN`) so PDs can stream the inventory across one or more fragments using reply code `0x88` (tentative).

Table 35: VCI Trust Anchor Status Request (osdp_PIVVCITASTATUS)

Size	Name	Meaning	Value
1	CMND	Command identifier.	`0xA6 (tentative)`

Table 36: VCI Trust Anchor Status Reply (osdp_PIVVCITASTATUSR)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	`0x88 (tentative)`
2	TOTAL	Length of the complete status payload, least-significant byte first.	`0x0000-0xFFFF`
2	OFFSET	Offset of this fragment within the status payload, least-significant byte first.	`0x0000-0xFFFF`
2	DATA_LEN	Length of the data block in this fragment, least-significant byte first.	`0x0000-0xFFFF`
1	Total Anchor Count	Number of cached trust anchors.	`0x00-0xFF`
2	Available Bytes	Remaining storage capacity for additional variable-length trust anchor records, least-significant byte first.	`0x0000-0xFFFF`
0-n	Anchor Records	Zero or more 12-byte inventory records. Each record contains: `TrustAnchorID` (1 byte for load/delete management), anchor selector `IIN` (8 bytes from the stored trust anchor record, not necessarily the secure CVC IIN), `AnchorLength` (2 bytes LSB first, maximum `0x0240`), and `Attributes` (1 byte, reserved). Records that span fragments continue in subsequent replies.	–

The PD first reports the total number of anchors and the remaining free bytes. *Available Bytes* is the amount of trust-anchor storage that remains for additional variable-length VCI trust anchor record data after accounting for currently stored anchors by their actual *Anchor Length* values. Each record thereafter lists a trust anchor by management ID, anchor selector IIN, anchor length, and reserved attributes (transmitted as zero). The IIN in this inventory is the IIN from the stored VCI trust anchor record and is not necessarily the secure messaging CVC IIN reported in osdp_CARDSTATUSR. Error responses follow the standard pattern: osdp_NAK function-not-supported when the command is not implemented, and osdp_PDERROR 0x0007 for multi-part errors.

Table 37: Trust Anchor Record Structure

Field	Size	Meaning	Value Range
Trust Anchor ID	1 byte	Identifier assigned by the ACU; reused to replace or delete an anchor.	0x00-0xFF
Issuer Identifier Number (IIN)	8 bytes	Anchor selector supplied in the VCI trust anchor record.	0x00-0xFF
Anchor Length	2 bytes (LSB first)	Total size of the stored variable-length VCI trust anchor record supplied for this anchor.	0x0000-0x0240
Attributes	1 byte	Reserved for future use; transmitted as zero.	0x00

A References

A.1 Normative References

- Security Industry Association, *Open Supervised Device Protocol (OSDP) v2.2*, 2020.
- National Institute of Standards and Technology, *NIST Special Publication 800-73-4*, *Interfaces for Personal Identity Verification (PIV) — Part 2: PIV Card Application Card Command Interface*, February 2015.
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- Internet Engineering Task Force, *RFC 2119*, *Key words for use in RFCs to Indicate Requirement Levels*, March 1997; and *RFC 8174*, *Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words*, May 2017.

A.2 Informative References

- National Institute of Standards and Technology, *NIST Internal Report 8347*, *Personal Identity Verification Cards, Version 2*, 2021.
- U.S. General Services Administration, *Federal Identity, Credential, and Access Management (FICAM) Architecture*, Revision 3, 2023.
- Open Physical, *High Speed Autonomous PIV: Concept Paper*, 2024.

B Sample osdp_PIVDATA Requests (Informative)

This appendix illustrates representative requests that conforming ACUs may issue using the clarified osdp_PIVDATA command. Object identifiers align with NIST SP 800-73-4 Part 2, Table 6. Each example shows the bytes that appear in the OSDP payload after the CMND field; the PD strips leading zeroes before constructing the ISO 7816-4 APDU.

Cardholder Unique Identifier (CHUID) Object identifier 0x5FC102, tag 0x00, offset 0x0000. The PD transmits GET DATA for tag 0x5FC102 and returns the complete CHUID BER-TLV so the ACU can validate the FASC-N, GUID, and expiration fields.

Card Capability Container (CCC) Object identifier 0x5FC107, tag 0x00, offset 0x0000. Use this request during device discovery to read feature flags (contact/contactless support, optional biometrics) prior to issuing PIV Mode updates.

Discovery Object Object identifier 0x00007E, tag 0x00, offset 0x0000. The PD forwards tag 0x7E with its nested data so the ACU can parse the application identifier (tag 0x4F) and PIN Usage Policy (tag 0x5F2F). To retrieve only the AID, repeat the command with tag 0x4F.

Biometric Information Templates (BIT) Group Template Object identifier 0x007F61, tag 0x00, offset 0x0000. Large responses for fingerprint or iris templates require multiple osdp_PIVDATAR fragments; the TOTAL/OFFSET/DATA_LEN fields support reconstruction.

Selective Tag Retrieval Object identifier 0x00007E, tag 0x4F, offset 0x0000. The PD returns the complete PIV Application Identifier BER-TLV nested inside the Discovery Object, including tag 0x4F, length, and value. Tags that occupy more than one byte (for example 0x7F61) are outside the scope of this selector and must be obtained by requesting tag 0x00.

C High-Speed Autonomous PIV Data Templates (Informative)

This appendix records the PIV Auto challenge, ISO/IEC 7816 `GENERAL AUTHENTICATE` templates, and fixture sources used by the demonstration vectors. The APDU examples are validated by the repository tests against the same helper code used by the simulator and replay tools.

C.1 Vector Sources

Table 38: PIV Auto Vector Sources

Source	Use	Notes
NIST SD 33 / IR 8347	Live-card APDU fixtures	Cards 2, 3, 4, 5, and 16 provide VCI contactless enrolment, OPACITY, and key-challenge transcripts.
NIST SM/VCI captures	CVC and trust anchors	Provides validator-ready 7F21, 5FC122, and 7F50 objects.
GSA FICAM card-builder	Local keys and certificates	Used only as reusable local test material for source-backed simulation.
Generated YubiKey PIV	Missing algorithm profiles	Reusable RSA-1024, RSA-3072, and P-384 test keys are committed as public fixtures and loaded through `ykmanpiv`; RSA-1024 is informative legacy test material only.

C.2 Supported Profiles

Table 39: PIV Auto Supported Profiles

Profile	`Alg`	`Key Ref`	PIV Auto	Coverage
RSA-1024	`06`	`9A, 9E`	No	Informative legacy/non-current test-only material.
RSA-2048	`07`	`9A, 9E`	Yes	NIST SD 33 cards 2, 3, 16; generated-card; GSA-backed simulation.
RSA-3072	`05`	`9A, 9E`	Yes	Committed YubiKey material; requires YubiKey firmware with RSA-3072 support.
ECC P-256	`11`	`9A, 9E`	Yes	NIST SD 33 cards 4 and 5; generated-card coverage.
ECC P-384	`14`	`9A, 9E`	Yes	Committed YubiKey material. NIST card 5 also provides 9C/9D APDU examples.

C.3 Fixed-Width KMAC Challenge Template

Table 40: PIV Auto KMAC Challenge Template

Field	Size	Encoding	Notes
Template label	14	ASCII plus version byte	`OSDP-PIV-AUTO` followed by 01.
Series Counter	4	LSB first	ACU-managed boot or epoch counter, for example 07000000.
Sequence Counter	4	LSB first	ACU-managed per-series counter, for example 2A000000.
FASC-N	25	Raw bytes	Credential FASC-N from CHUID.
UUID	16	Raw bytes	Credential UUID. Successful PIV Auto examples use the credential UUID; unavailable UUID is a failure condition.
PIV Key Reference	1	Raw byte	Challenged key reference, normally 9A or 9E.
PIV Algorithm	1	Raw byte	Algorithm identifier: 05, 06, 07, 11, or 14.
Supplemental Entropy	32	Raw bytes	Fixed-width ACU entropy field; all zeroes are valid for test vectors only.

The challenge digest is produced in two steps. First, derive `PIV-AUTO-KDK=KMAC256(key=S-ENC||S-MAC1||S-MAC2,data=empty,mac_len=32,custom="OSDP-PIV-AUTO-KDK-v1")`. Second, derive the challenge as `KMAC256(key=PIV-AUTO-KDK,data=fixed-widthtemplate,mac_len=L,custom="OSDP-PIV-AUTO-CHALLENGE-v1")`. `L` is 32 bytes for SHA-256 profiles and 48 bytes for SHA-384 profiles. The supplemental entropy field is always present and fixed at 32 bytes; all zeroes are valid for test vectors but not recommended for deployments.

C.4 Dynamic Authentication Templates

C.4.1 RSA Key Challenge Input

Table 41: RSA Dynamic Authentication Template

Tag / Bytes	Length	Name	Value
7C	Variable	Dynamic Authentication Template	Outer template for the PIV GENERAL AUTHENTICATE command data.
82	00	Response request	Empty value requesting the card response in tag 82.
81	Modulus length	Challenge block	EMSA-PKCS1-v1_5 encoded digest block with the same length as the RSA modulus.
0001FF...FF00	Variable	Padding prefix	EMSA-PKCS1-v1_5 block prefix before the digest information.
30...	Variable	DigestInfo	ASN.1 DigestInfo for SHA-256 or SHA-384 followed by the KMAC-derived digest; SHA-256 uses prefix 3031300D060960864801650304020105000420 and SHA-384 uses prefix 3041300D060960864801650304020205000430.

Table 42: RSA Dynamic Authentication Examples

Profile	Alg	Key Ref	Tag 81	Response 82	Command APDU
RSA-1024	06	9A/9E	128 bytes	128 bytes	Informative legacy/non-current test-only APDU.
RSA-2048	07	9A/9E	256 bytes	256 bytes	274-byte extended APDU.
RSA-3072	05	9A/9E	384 bytes	384 bytes	402-byte extended APDU.

C.4.2 ECC Signature Input

Table 43: ECC Dynamic Authentication Template

Tag / Bytes	Length	Name	Value
7C	Variable	Dynamic Authentication Template	Outer template for the PIV GENERAL AUTHENTICATE command data.
82	00	Response request	Empty value requesting the card response in tag 82.
81	Digest length	Challenge digest	KMAC-derived digest supplied directly to the ECC signing operation.
Digest bytes	32 or 48 bytes	Digest value	SHA-256-width digest for P-256; SHA-384-width digest for P-384.

Table 44: ECC Dynamic Authentication Examples

Profile	Alg	Key Ref	Tag 81	Response 82	Command APDU
ECC P-256	11	9A/9E	32 bytes	DER ECDSA signature, about 70-72 bytes	44-byte short APDU.
ECC P-384	14	9A/9E	48 bytes	DER ECDSA signature, about 102-104 bytes	60-byte short APDU.

C.4.3 Worked Dynamic Authentication Examples

Worked examples are generated from `test-vectors/piv-auto-demo/generated/piv-auto-simulation-report.json`.

Table 45: PIV Auto Worked Example: PIV Authentication RSA-1024

Field	Value
Profile	9a-rsa1024
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9a-rsa1024. Informative legacy/non-current test-only coverage.
Algorithm ID	0x06
Key Reference	0x9A
KMAC Challenge Digest	5B2FC08860E3999EEF8F6CF5D2B0C9AE05CBB28A5B423BFE4F646F6F47FCA4F2

Field	Value
Tag 81 Challenge Input	0001FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFF003031300D060960864801650304020105000420 5B2FC08860E3999EEF8F6CF5D2B0C9AE05CBB28A5B423BFE4F646F6F47FCA4F2
GENERAL AUTHENTICATE APDU	0087069A887C818582008181800001FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF003031300D0609 608648016503040201050004205B2FC08860E3999EEF8F6CF5D2B0C9AE05CBB2 8A5B423BFE4F646F6F47FCA4F200
Dynamic Authentication Response	7C81838281806FE396FF8D6F618EDFF35CC600ECE5DFECB74C66D6FE90352012 71D2230484F989A70619293F8450AE67AF864C8C1F35AFBDC8D531D28AB6E3E8 335588E123C224F6737807DCD3DCF78EDD6853FE216C2FEBF78DFCD724265095 4781DEF3FBBC6C772E6CB65D1135051CD116FBAB39E3BDFBACA0A9949C6BE518 9290BA83498A9000
Response Tag 82 Length	128 bytes

Table 46: PIV Auto Worked Example: Card Authentication RSA-1024

Field	Value
Profile	9e-rsa1024
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9e-rsa1024. Informative legacy/non-current test-only coverage.
Algorithm ID	0x06
Key Reference	0x9E
KMAC Challenge Digest	DB6851587148B203545E4306315583B0BB0CB88B8534200FDE187CF4F8F37D4F
Tag 81 Challenge Input	0001FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFF003031300D060960864801650304020105000420 DB6851587148B203545E4306315583B0BB0CB88B8534200FDE187CF4F8F37D4F
GENERAL AUTHENTICATE APDU	0087069E887C818582008181800001FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF003031300D0609 608648016503040201050004 20DB6851587148B203545E4306315583B0BB0CB8 8B8534200FDE187CF4F8F37D4F00
Dynamic Authentication Response	7C818382818023F95C29066CB6513196C034958EF17FF08135027485AC6C91EE 230226E7A81F446CB1CAD207C2B4D8F32C6BFD78824297270DAD5C66388208B8 68C790120CCC247E134C193AA7DE60CB6E66A5AC104867B0BEAB0468C8558B26 E45FED226B4DA7F0EF9F8339A531CEDF354BEA70094F01E01267D0EEC359B476 8FA360469C879000
Response Tag 82 Length	128 bytes

Table 47: PIV Auto Worked Example: PIV Authentication RSA-2048

Field	Value
Profile	9a-rsa2048
Source	Committed RSA-2048 card-authentication key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x07
Key Reference	0x9A
KMAC Challenge Digest	E5D733C7FBBA8344701EE8909F1E6642E040CCF3265E0D5D28A26D73C344C19C
Tag 81 Challenge Input	0001FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFF003031300D060960864801650304020105000420 E5D733C7FBBA8344701EE8909F1E6642E040CCF3265E0D5D28A26D73C344C19C
GENERAL AUTHENTICATE APDU	0087079A00010A7C8201068200818201000001FFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF003031 300D060960864801650304020105000420E5D733C7FBBA8344701EE8909F1E66 42E040CCF3265E0D5D28A26D73C344C19C00
Dynamic Authentication Response	7C8201048282010051535 6AB0012D988701AC373B029D86540C4DF473B031EDB 5ADFDB84BC214A45497F6CEE5FFB15ECB0CD89B9F351EF5006AB20105EF289EE 041F427802867BA48253B6B90F0BD6F77637B14DD36B32BE89F8BD12F2D9F0E5 B408C6C452959F233EDB9FA0A96B2808711FB1B671A683460F5FD62847AC18E5 717B7D008EDE5FB2AEDFFF44EDD90B2E2DB595456BA7D42644C57C2267C4B996 1424839E352FE64EB5F344CBB12BC630FFCDE943055A80375CF6AEA11BAF6AEC 2916D5A1B6D769CC5F935519634C2504A2CA4916368DAAA887DC61A514AC630D B093CD2AC3608CECD355A63DCA02E35FBF2CAEDA34ED51C7F60C69EFE7CCCFDA 9115A93950016990 9000
Response Tag 82 Length	256 bytes

Table 48: PIV Auto Worked Example: Card Authentication RSA-2048

Field	Value
Profile	`9e-rsa2048`
Source	Committed RSA-2048 card-authentication key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	`0x07`
Key Reference	`0x9E`
KMAC Challenge Digest	`0864C776F2374124D3E63F0B0B29FC1C5F0E8FF8BB1FA80E2723293B86A0158E`
Tag 81 Challenge Input	0001FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFF003031300D060960864801650304020105000420 0864C776F2374124D3E63F0B0B29FC1C5F0E8FF8BB1FA80E2723293B86A0158E
GENERAL AUTHENTICATE APDU	0087079E00010A7C8201068200818201000001FFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF003031 300D0609608648016503040201050004200864C776F2374124D3E63F0B0B29FC 1C5F0E8FF8BB1FA80E2723293B86A0158E00
Dynamic Authentication Response	7C82010482820100 9BB1AA3875DB4E41107DD91CEAAC36FE4CE5F969E1F7B9B1 99DFAB38CFA1EE36FBF197CD2777C4A6CE082623C9F36A9DC2514FE54DD12934 AFCAE16DFE14FFCFC0F58E0A0F46567444968F1DE78B496B4222DB913A87B850 139B1E2A730124C4ED169919F1EB5FA3E15617E89BB9CED8FA996A63E126C239 EB6D142A5746FBEC3DFA3B287F944EE45586BA5AEB64087BDA0661DF7254C42D 9768141F7243382D61CC43A8CDBE4ADA849E05F013D947DF7FA608C38ABA3841 B038AA69D77461B00053F59C4D34906BFBFD470A277D0722AAECD83117DE906B FCD36A16EFB00B3EACD4414F48528CCA4BE57826B4312424A65DABAD6FA1ECA3 DAF55F6E6E1022C89000
Response Tag 82 Length	256 bytes

Table 49: PIV Auto Worked Example: PIV Authentication RSA-3072

Field	Value
Profile	`9a-rsa3072`

Source	`Committed YubiKey loader material in`
	`test-vectors/piv-auto-yubikey-material/9a-rsa3072.`
	`Generated-card coverage.`

Algorithm ID	`0x05`
Key Reference	`0x9A`
KMAC Challenge Digest	`1503FD1E89AA91A5B37E6872AC3846F654B7B1D099B2F3AEB597472DFE38E714` `42816BC43454622219F61A3C6DAFDB4F`

Tag 81 Challenge Input	`0001FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF00304130`
	`0D0609608648016503040202050004301503FD1E89AA91A5B37E6872AC3846F6`
	`54B7B1D099B2F3AEB597472DFE38E71442816BC43454622219F61A3C6DAFDB4F`

GENERAL AUTHENTICATE APDU	`0087059A00018A7C8201868200818201800001FFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF`
	`FFFFFFFFFFFFFFFFFFFFFFFFFF003041300D060960864801650304020205000 4`
	`301503FD1E89AA91A5B37E6872AC3846F654B7B1D099B2F3AEB597472DFE38E7`
	`1442816BC43454622219F61A3C6DAFDB4F00`

Field	Value
Dynamic Authentication Response	7C8201848282018019C03ACCABE74C9259FA5A2267BC4F1401AB352F210BEFE0 F74B8B74F25C82E0A83E1635C19C341CBD97D71D856BF132A2C3CDD223AC7D37 451E9083A0C5FDB8013577741520A8A2375896E3839252F10DD8826A00A065F0 6154EE7A51703D65CC00AAA66E28BE5683085B4045882AA17B925457D3B6B0BC 0C4581D10ED4653A1D4713E4FB5A4C72054789C9A4C1E6BC2B6B497228A9C765 365040D00C139671175C262BA2E6D6064F1F84AAC0D5279B91F9FDE6D6B02B24 23C7CC3FA7408E80A8040B92B819F80D6AD23B86C8F339D50F332802CBED0511 215B03DF9496DD64F77ADBE4AE844F0D708F230E6F06911DADCA6888B9C80C94 A16FCB0DE7467B70766668CDEA0053289295256F8BEDC693F058F6244BB6B315 C4C9F07D89432A479AF72C666181C7B47CAA72C7ABE8B323658BD681FA10A1E8 8E2AFAB03BFDE9B571CA6FEDA05A24CC6436061D633691822FDEDD3DAE4E130A 1CCE50E65FDFC5EA451B7AA8F83D4385D40E005E368D064E105D6F5325B36D95 5F3D88B5BEF840789000
Response Tag 82 Length	384 bytes

Table 50: PIV Auto Worked Example: Card Authentication RSA-3072

Field	Value
Profile	9e-rsa3072
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9e-rsa3072. Generated-card coverage.
Algorithm ID	0x05
Key Reference	0x9E
KMAC Challenge Digest	4C98712B25DB7F477096FA3D0AC93FA2CA6BBE89CB291AC26EDF0D8312CA61D0 CA238F1656247B2DF38E8B72D9AF7F5E
Tag 81 Challenge Input	0001FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF00304130 0D060960864801650304020205000430 4C98712B25DB7F477096FA3D0AC93FA2 CA6BBE89CB291AC26EDF0D8312CA61D0CA238F1656247B2DF38E8B72D9AF7F5E

Field	Value
GENERAL AUTHENTICATE APDU	0087059E00018A7C8201868200818201800001FFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFFFF003041300D0609608648016503040202050004 304C98712B25DB7F477096FA3D0AC93FA2CA6BBE89CB291AC26EDF0D8312CA61 D0CA238F1656247B2DF38E8B72D9AF7F5E00
Dynamic Authentication Response	7C820184828201807 4FA8827F5C2CFDAA5718679727015CD5A928828997F2F14 23D033DE8AADEB3E86431170A714F5155B830F65F3F854120B16BC50F701C0EE A0278DB7EC2D7D6D587BEDB32F5E34EF1421E2C42543FF7DEEF7C5A41A01275B 0BC126AF1208BBBD1E2C65149BC74C9E2CD00D822081B22E5CCB5090D2BD3970 CA309D2124D79E9A28978D5DDF26E3FA8E37C360521C5BBE1AB27FACC24FF4C5 9AB46B7D571777B22E3E4981593D9F689DA3D33F3DE7C385C4315A7F154BE461 968EA05A3315C4424717DD6F3CBF7FB94A135AE33573E58A18185F7E25771E6C 4AB2B89D50007C7A60E6D65CF3BE58F8901B25B867D0FF97981BCBEE2BFCF5E5 7CEB5A81F40EEB44E52A0BD7F538F932AB4F1732828F2E95C4F203967ADF5983 A285A127E6C0D7C2048261290B6B58D8110056DFE65547A7C1BDF78C6FEA62A8 E07EC932AC6B58AD13C9423AAD5815C5251BCECEBBDB181F0B1C362163373463 5FD48CD1AD16798C9B67CDF2BAB290DE0EA0BC5BA54E41736BA4885AF72D729A 5168DC09DDD4B6629000
Response Tag 82 Length	384 bytes

Table 51: PIV Auto Worked Example: PIV Authentication ECC P-256

Field	Value
Profile	9a-ecp256
Source	Committed P-256 signing key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x11
Key Reference	0x9A
KMAC Challenge Digest	A3C81D3205FFB28A2AAAB0EDC21E8B3BEF5BD5F685FF5C686502CA5609B45851
Tag 81 Challenge Input	A3C81D3205FFB28A2AAAB0EDC21E8B3BEF5BD5F685FF5C686502CA5609B45851
GENERAL AUTHENTICATE APDU	0087119A267C2482008120A3C81D3205FFB28A2AAAB0EDC21E8B3BEF5BD5F685 FF5C686502CA5609B4585100

Field	Value
Dynamic Authentication Response	7C4A824830460221009D25A44E56572983819019DF26A2720E462A11D2CB3CD8 6A9142DEF4BCC1271B022100D084982BB9D3E2D3E6B0965956C71DBA0FE44FE3 28C0A2BD72951DFB4D9296FE9000
Response Tag 82 Length	72 bytes

Table 52: PIV Auto Worked Example: Card Authentication ECC P-256

Field	Value
Profile	9e-ecp256
Source	Committed P-256 signing key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x11
Key Reference	0x9E
KMAC Challenge Digest	965AF088B02DF65E7C299ABF53A5AB2DCBB3C789458DE8D0C819B19A9266658D
Tag 81 Challenge Input	965AF088B02DF65E7C299ABF53A5AB2DCBB3C789458DE8D0C819B19A9266658D
GENERAL AUTHENTICATE APDU	0087119E267C2482008120965AF088B02DF65E7C299ABF53A5AB2DCBB3C78945 8DE8D0C819B19A9266658D00
Dynamic Authentication Response	7C4A8248304602210 08B204FA48FAA27A57D6E90F985199F284D2A8E0D0F2E05 1BBFBCACBA824619B9022100C6E04880D322DE7D10823EC78C8EA4CCB69FE394 F694F1D12C797FA7BCAD76959000
Response Tag 82 Length	72 bytes

Table 53: PIV Auto Worked Example: PIV Authentication ECC P-384

Field	Value
Profile	9a-ecp384
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9a-ecp384. Generated-card coverage.
Algorithm ID	0x14
Key Reference	0x9A
KMAC Challenge Digest	C813683447986D513FC17CED5DB4D335BFBEAFB1AF9E5200C6B7B1C1E3535D70 0C2E05F9011717E153998832B7588D28
Tag 81 Challenge Input	C813683447986D513FC17CED5DB4D335BFBEAFB1AF9E5200C6B7B1C1E3535D70 0C2E05F9011717E153998832B7588D28
GENERAL AUTHENTICATE APDU	0087149A367C3482008130C813683447986D513FC17CED5DB4D335BFBEAFB1AF 9E5200C6B7B1C1E3535D700C2E05F9011717E153998832B7588D2800

Field	Value
Dynamic Authentication Response	7C6A82683066023100D88ABABB9893D88341CFA2F1AB6E3D35E4AF02F4F8D085 B5A3DE63574DCF694741D869066B8E6297429120614DCC9CE40231009A3F203B B5EA98839008B0D25F98C98DE8E87622F61279B2F2B29343912999A36311102B 310F9985844BCC248350377B9000
Response Tag 82 Length	104 bytes

Table 54: PIV Auto Worked Example: Card Authentication ECC P-384

Field	Value
Profile	9e-ecp384
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9e-ecp384. Generated-card coverage.
Algorithm ID	0x14
Key Reference	0x9E
KMAC Challenge Digest	F3480C6C1DAD008D3E14D0C815D381F01420E9D3402A175BED097C6949E4BA44 7FA0A11745CE4D054A95B10A9D5CC689
Tag 81 Challenge Input	F3480C6C1DAD008D3E14D0C815D381F01420E9D3402A175BED097C6949E4BA44 7FA0A11745CE4D054A95B10A9D5CC689
GENERAL AUTHENTICATE APDU	0087149E367C3482008130F3480C6C1DAD008D3E14D0C815D381F01420E9D340 2A175BED097C6949E4BA447FA0A11745CE4D054A95B10A9D5CC68900
Dynamic Authentication Response	7C6982673065023034C6B2659FBE26A78D2615F8C0EBF2E9674292D9DE3CE8A3 1D9DA297B5AE236AC886E9E9C288D42474564BF0C46AE40C023100CCBD59A047 D73F9FD58F60218FBCB4F99B18EBD7EB70EFE5CF64D5EB1A65F1676A867E1B1E 5EB4CF9F34B44BE905D29C9000
Response Tag 82 Length	103 bytes

C.5 APDUs

Table 55: PIV Auto APDU Reference: Card Preparation

Operation	Command APDU	Success Response	Purpose
Select PIV Application	00A404000BA000000308000010000100	9000	Selects the PIV application before reading objects or authenticating.
Get Discovery Object	00CB3FFF035C017E	7E...9000	Reads the Discovery Object so the PD can determine application and PIN/VCI policy.

Operation	Command APDU	Success Response	Purpose
Verify PIN (optional)	00200080083132333435 36FFFF	9000	Verifies example PIN 123456 padded with FF; failure responses include 63CX and 6983.

Table 56: PIV Auto APDU Reference: GENERAL AUTHENTICATE Command

Field	Encoding	Source	Notes
CLA	00	ISO/IEC 7816-4	Use 0C when a credential secure channel is active.
INS	87	ISO/IEC 7816-4	GENERAL AUTHENTICATE instruction.
P1	Algorithm ID	PIV profile	05, 07, 11, or 14 for supported PIV Auto profiles; 06 appears only in informative RSA-1024 legacy examples.
P2	Key Reference	PIV profile	9A or 9E for PIV Auto.
Lc	Short or extended	Command length	Short APDUs use one length byte; extended APDUs use 00 followed by two length bytes.
Command data	7C...820081......	Dynamic Authentication Template	Tag 82 is empty to request a response; tag 81 carries the exact challenge input.
Le	00	Response request	Requests the maximum response supported by the card interface.

Table 57: PIV Auto APDU Reference: GENERAL AUTHENTICATE Response

Case	Response APDU	PD Handling	OSDP Payload
Complete response	7C...82...9000	Strip only the ISO status word.	Carry the complete 7C Dynamic Authentication Template in osdp_CRAUTHR or osdp_PIVSTATUS.
Additional response bytes	61XX	Issue GET RESPONSE with Le XX.	Concatenate returned data before reporting the final 7C template.
GET RESPONSE command	00C00000XX	Retrieve remaining card response bytes.	Not exposed directly to the ACU.

Case	Response APDU	PD Handling	OSDP Payload
Credential error	69.. or 6A..	Preserve command failure detail.	Report via osdp_PDERROR when available; do not treat as a valid signed response.

Table 58: PIV Auto APDU Reference: OPACITY / VCI Context

Operation	Command Shape	Key Ref	Relationship to PIV Auto
OPACITY establish	0087[27/2E]04[Lc]7C...81...8200	04	Establishes VCI secure messaging and returns cryptogram/CVC evidence.
PIV Auto challenge	0087[alg][9A/9E][Lc]7C...820081......00	9A/9E	Performs the autonomous key challenge after card preparation and any required VCI setup.
Response handling	7C...82...9000	9A/9E	The PD reports the returned signed response bytes exactly, excluding only SW1/SW2.

D Deprecated osdp_GENAUTH

The osdp_GENAUTH command is deprecated and expected to be removed in a future revision of this specification. Using a PIV credential to decrypt or process arbitrary data is unsafe and unnecessary for credential verification; implementations **SHOULD** use osdp_CRAUTH flows instead. Controllers and PDs **SHOULD** plan migrations away from osdp_GENAUTH.